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Periodic Boundary Conditions in Generalized Beam Theory: A Shear-Deformable GBT–Ritz Formulation for the Buckling of Omega-Stringer-Stiffened Composite Panels

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Abstract: Stringer-stiffened composite panels in aircraft covers carry many identical stringers, so that their buckling behaviour is commonly analysed on one representative cell whose side edges satisfy periodic boundary conditions. Closed-form models with periodic conditions rely on idealisations of the skin–stringer interaction, while shell finite element models with periodic constraint equations are accurate but expensive and give no insight into the deformation mechanisms. Generalized Beam Theory (GBT) provides such insight at low cost, but its cross-section analysis has been developed for members with free longitudinal edges or closed contours, and periodic side edges have not been treated. This study combines GBT and the Ritz method into a shear-deformable GBT–Ritz formulation for omega-stringer-stiffened composite panels in which the periodic condition is built into every family of deformation modes: the conventional modes through constraint rows that are projected out family by family, the warping functions through a constrained Vlasov least-squares problem, and the shear and transverse-shear modes through merging of the nodal functions of the two side edges. Every mode, and therefore every truncated mode basis, satisfies the periodic condition exactly. The walls are laminated Reissner–Mindlin plates with full membrane–bending and anisotropic coupling, the stringer is tied to the skin with its true offset, the pre-buckling state is the first-order solution under the actual end loads, and the amplitude functions are cubic B-splines, which cover simply supported and clamped loaded edges. The modal tensors, equilibrium equations and boundary conditions are derived in the notation of the shear-deformable GBT of Silvestre and Camotim, and the formulation reduces to the classical GBT buckling equations for uncoupled walls under uniform compression. For 30 panels with simply supported loaded edges (five lay-ups, two stringer pitches, three aspect ratios), the critical loads agree with Abaqus shell models with periodic constraint equations within 0.57 % on average and 1.25 % at most; for the 30 panels with clamped loaded edges, they agree with an independent spline finite strip model within 0.30 % on average and 1.17 % at most. An energy-based mode decomposition identifies all critical modes as local skin buckling with global and distortional contributions below 4 %, at a computing time of a few seconds per case.

Keywords: generalized beam theory; Ritz method; periodic boundary conditions; stiffened panels; omega stringer; laminated composites; buckling

1. Introduction

Stringer-stiffened panels are the standard construction of aircraft fuselage and wing covers, and their buckling behaviour is often the governing design criterion of weight-critical laminated composite structures [1,2]. Among the stiffener shapes, the omega (hat) stringer is attractive because its closed cell provides high bending and torsional stiffness, a wide bonding area and good damage tolerance [3,4]. A typical cover carries many identical stringers. Its buckling behaviour away from the free or supported longitudinal edges is therefore represented by one repeating cell, provided that the side edges of this cell are given the conditions that the neighbouring cells impose, namely periodic boundary conditions [5–7].

The treatment of the repeating cell is a classical topic of the analysis of stiffened panels. In exact and finite strip methods for prismatic plate assemblies, the infinitely wide panel was analysed early through its repeating portion: the exact strip programs VIPASA and VICONOPT treat infinitely wide or long panels by repeating a portion and its supports [8,9], and the finite strip method in general handles such assemblies with a small number of strips [10,11]. The underlying principle is the theory of waves in periodic structures (Bloch–Floquet theory), by which the response of an infinite periodic structure follows from one cell with phase-related boundary conditions [12].

At the Institute for Lightweight Engineering and Structural Mechanics (LSM), a series of closed-form and semi-analytical models for stringer-stiffened composite panels with periodic boundary conditions has been developed. Mittelstedt derived closed-form buckling loads of blade-stringer-stiffened composite plates considering periodic boundary conditions [5] and extended the approach to the stiffener–plate interaction in

local buckling [6]. For omega stringers, Mittelstedt and Beerhorst presented a closed-form buckling analysis [3], and Mittelstedt and Schröder treated the local postbuckling of hat-stringer-stiffened plates [13]. Schilling and Mittelstedt developed a closed-form approximate solution for the local buckling of omega-stringer-stiffened panels [4], a column-buckling model with periodic boundary conditions [7], and models for shear buckling, compression–shear interaction and local postbuckling [14–16]. Most recently, El Yaakoubi-Mesbah and Mittelstedt presented a closed-form analytical solution for the local buckling of omega-stringer-stiffened composite panels under compression [17], a computational model for local buckling [18] and a closed-form postbuckling analysis including the stringer-foot effects [19]; the stability of stiffened laminates is summarised in [20]. These models are efficient and transparent, but they rest on idealisations: the stringer is replaced by elastic restraints, the skin is reduced to one or a few plate strips with assumed shape functions, local and global buckling are usually treated by separate models, and the pre-buckling stress is taken as uniform. The membrane–bending coupling of unsymmetric laminates is typically included through a reduced bending stiffness [21]. The physics-informed neural network of the authors for the local buckling of omega-stringer-stiffened panels [22] was built on such a reduced energy model. Shell finite element models with periodic constraint equations between the nodes of the two side edges, on the other hand, are accurate and general, but each design requires a new mesh and a large eigenvalue analysis, and the buckling mode is obtained only as a nodal field, without a decomposition into mechanically meaningful deformation mechanisms.

Generalized Beam Theory (GBT), originated by Schardt [23,24] and developed further by Davies and Leach [25], describes the deformation of a prismatic thin-walled member as a superposition of cross-section deformation modes whose amplitudes vary along the member axis. A cross-section analysis determines the modes and the associated modal matrices, and a member analysis determines the amplitude functions. The modal decomposition separates global, distortional, local and shear-related deformations and has made GBT one of the most transparent tools for coupled instabilities of thin-walled members, as summarised in the state-of-the-art reports by Camotim et al. [26] and Mittelstedt [27,28]. Silvestre and Camotim formulated first- and second-order GBT for arbitrary orthotropic materials [29,30], and later shear-deformable GBT for composite members and its vibration counterpart [31,32]. Dinis et al. extended the cross-section analysis to arbitrarily branched open sections [33]. Gonçalves et al. introduced a new approach to the calculation of the cross-section deformation modes [34], and Bebiano et al. rationalised the cross-section analysis into natural, shear and transverse-extension mode families [35], together with the shear deformation modes [36], implemented in the code GBTUL 2.0 [37]. Closed sections were treated, among others, for circular cylindrical shells and tubes [38] and for regular polygonal tubes [39]. For laminated composites, GBT has been applied to FRP open-section columns [40] and to laminated CFRP cylindrical stiffened panels [41], and Taig and Ranzi formulated GBT for stiffened sections [42].

The Ritz method provides the complementary tool for the member analysis: the continuous buckling problem is replaced by a finite-dimensional one over admissible functions that satisfy the essential boundary conditions [43], and its efficiency depends on a basis that reflects the expected solution [44]. Outside GBT, Kühn et al. used such variational discrete-plate models for the local buckling of shear-deformable laminated beams [45]. Kharghani and Mittelstedt combined GBT and the Ritz method into a GBT–Ritz approach: for global, local and distortional buckling of shear-deformable composite beams with open cross-sections [46], for composite box girders [47], and for thin-walled composite beams with unsymmetric layups, for which the membrane–bending coupling is consistently included [48,49]. In this approach GBT defines the kinematics through the modes, and the Ritz method determines how the modal amplitudes vary along the member.

Two gaps motivate the present study. The first gap lies in GBT. GBT has been developed for members: open sections with free longitudinal edges, branched sections, and closed contours. The skin of a stiffened panel cell, however, ends at two side edges that are neither free nor closed: the displacement field at one side edge must equal that at the other. This condition has not been formulated in GBT, and it cannot simply be imposed afterwards on a set of classical modes. Classical mode generation starts from a few natural nodes, whereas the periodic condition couples the two ends of the open contour; a truncated set of classical modes then contains too few modes that satisfy it, and the rigid-body and warping modes of the open cell are not admissible for the periodic cell. In addition, the omega stringer forms closed cells with the skin, so that Vlasov's condition of null membrane shear strain cannot be fulfilled by the warping functions, the stringer is offset from the skin by the half thicknesses, and the pre-buckling state is not uniform near the loaded edges, where the load enters the stringer through the eccentric foot. The second gap lies in the treatment of periodic conditions. The closed-form models are efficient, but they rely on idealisations of the skin–stringer interaction, separate local and global buckling, and do not account for the non-uniform pre-buckling state; shell finite element models do not have these restrictions, but they are expensive for parametric studies and give no modal insight; exact and finite strip methods treat the repeating portion efficiently, but they are

formulated in nodal degrees of freedom and do not provide the decomposition into global, distortional and local mechanisms.

GBT and periodic boundary conditions complement each other. The periodic cell is prismatic, so that the member analysis of GBT applies directly; the modal decomposition identifies the buckling mechanisms of the cell; and the reduced number of generalised coordinates makes parametric studies cheap. The present study combines them. Its novelty is the formulation of the periodic condition within the cross-section analysis of GBT, such that every deformation mode, and therefore every combination and truncation of the modes, satisfies it exactly, together with a shear-deformable GBT–Ritz member analysis of omega-stringer-stiffened panels with symmetric and unsymmetric laminates. The formulation is derived in the notation of the shear-deformable GBT of Silvestre and Camotim [31]: the modal tensors, equilibrium equations and boundary conditions are given explicitly, extended by the membrane terms that the closed contours of the cell require, and solved by the GBT–Ritz approach [49]; the formulation reduces to the classical GBT buckling equations for uncoupled walls under uniform compression. The formulation is verified against Abaqus shell models with periodic constraint equations for simply supported loaded edges and against an independent spline finite strip model for clamped loaded edges.

The paper is organised as follows. Section 2 develops the formulation: the cross-section analysis of the periodic cell, the periodic boundary conditions, the member analysis by the Ritz method and the mode decomposition. Section 3 presents and discusses the results, and Section 4 summarises the conclusions.

2. Theory and formulations

The formulation follows the two stages of GBT [24,29,35]: a cross-section analysis, which determines the deformation modes and the modal mechanical properties (tensors), and a member analysis, which determines the modal amplitude functions. The notation is that of the shear-deformable GBT of Silvestre and Camotim [31], and the member analysis follows the GBT–Ritz approach [46,49]. Three extensions are required for the panel cell: (i) the deformation modes must satisfy the periodic boundary conditions of the cell (Section 2.2), (ii) Vlasov's assumptions of null membrane shear strain and transverse extension cannot be imposed on the conventional modes of a cell that contains closed contours, and (iii) the walls are shear-deformable (Reissner–Mindlin) laminates. Each of them is introduced such that the classical GBT equations are recovered as a special case (Section 2.3.6).

2.1. Cross-section analysis

2.1.1. Periodic cell, discretisation and displacement field

The panel is the periodic cell of an infinitely wide panel with identical omega stringers at the pitch b_s (Figure 1a). The cell contains two stringers and has the width $W = 2b_s$ and the length $L = \beta b_s$, with the aspect ratio β. The cross-section axes are X across the panel and Y normal to the skin, towards the stringers; x is the member axis along the stringers. The skin occupies $-W \leq X \leq 0$, and its side edges $X = -W$ and $X = 0$ lie in the middle of a skin bay. As in Silvestre and Camotim [31], every wall has the local axes (x, s, z), with s along the wall mid-line and z through the thickness, and the displacement components u, v and w (Figure 1b). The mid-line is divided into walls (skin, feet, webs, crowns) and the walls into segments of width at most $b_s/25$, with at least two segments per wall; the end points of the walls are the natural nodes and the remaining segment end points the intermediate nodes [24,35]. For $b_s = 150$ mm the cell has $N = 92$ nodes, 19 walls and 93 segments. The two side nodes of the skin are denoted s_L ($X = -W$) and s_R ($X = 0$).

The displacement field is the modal representation of shear-deformable GBT [31],

$$u(x,s) = u_k(s)\,\phi_{k,x}(x) + u_j(s)\,\varphi_j(x), \qquad v(x,s) = v_k(s)\,\phi_k(x), \qquad w(x,s) = w_k(s)\,\phi_k(x), \quad (1)$$

extended here by the transverse shear rotations of the Reissner–Mindlin walls,

$$\beta_x(x,s) = \xi_r(s)\,\chi_r(x), \qquad \beta_s(x,s) = \zeta_r(s)\,\chi_r(x), \quad (2)$$

where the summation convention applies to the mode indices; k (and i for virtual quantities) denotes the conventional modes, j (and h) the shear modes, and r the transverse shear (Mindlin) modes. $u_k(s)$, $v_k(s)$ and $w_k(s)$ are the displacement profiles of conventional mode k and $\phi_k(x)$ its amplitude function; $u_j(s)$ is the warping profile of shear mode j and $\varphi_j(x)$ its amplitude function; $\xi_r(s)$ and $\zeta_r(s)$ are the rotation

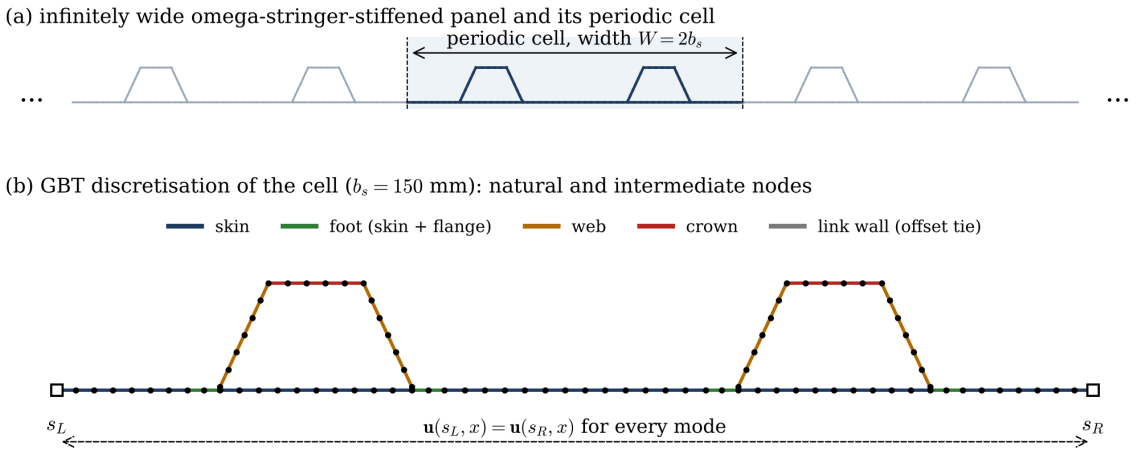


Figure 1. (a) Infinitely wide omega-stringer-stiffened panel and its periodic cell of width $W = 2b_s$. (b) GBT discretisation of the cell for $b_s = 150$ mm with natural and intermediate nodes; skin, feet (skin and flange as one laminate), webs, crowns and link walls of the offset tie; s_L and s_R are the side nodes on which the periodic condition acts.

profiles of Mindlin mode r and $\chi_r(x)$ its amplitude function. The uniform axial extension is represented by a shear-type mode with $u_j = 1$, i.e. by $\varphi_j = \phi_{1,x}$ of the axial extension mode of classical GBT.

2.1.2. Strain–displacement relations

The rotations of the wall normal are $\phi_x = -w_{,x} + \beta_x$ and $\phi_s = -w_{,s} + \beta_s$, and the linear strains at the distance z from the mid-surface follow as

$$\varepsilon_{xx} = u_{,x} + z\,\kappa_{xx}, \qquad \varepsilon_{ss} = v_{,s} + z\,\kappa_{ss}, \qquad \gamma_{xs} = u_{,s} + v_{,x} + z\,\kappa_{xs}, \qquad \gamma_{xz} = \beta_x, \qquad \gamma_{sz} = \beta_s, \tag{3}$$

with the curvatures $\kappa_{xx} = -w_{,xx} + \beta_{x,x}$, $\kappa_{ss} = -w_{,ss} + \beta_{s,s}$ and $\kappa_{xs} = -2w_{,xs} + \beta_{x,s} + \beta_{s,x}$. Inserting Eqs. (1) and (2) gives

$$\begin{aligned} \varepsilon_{xx} &= (u_k - z\,w_k)\,\phi_{k,xx} + u_j\,\varphi_{j,x} + z\,\xi_r\,\chi_{r,x}, \qquad \varepsilon_{ss} = \left(v_{k,s} - z\,w_{k,ss}\right)\phi_k + z\,\zeta_{r,s}\,\chi_r, \\ \gamma_{xs} &= \left(\gamma_k - 2z\,w_{k,s}\right)\phi_{k,x} + u_{j,s}\,\varphi_j + z\left(\xi_{r,s}\,\chi_r + \zeta_r\,\chi_{r,x}\right), \qquad \gamma_{xz} = \xi_r\,\chi_r, \qquad \gamma_{sz} = \zeta_r\,\chi_r, \end{aligned} \tag{4}$$

where $\gamma_k = u_{k,s} + v_k$ is the membrane shear strain profile of conventional mode k. In the GBT of Silvestre and Camotim [31], the conventional modes satisfy Vlasov's assumptions, $\gamma_k = 0$ and $v_{k,s} = 0$, and Eq. (4) reduces to their kinematic relations ($\xi_r = \zeta_r = 0$). Here these assumptions are not imposed, for two reasons: the transverse extension modes ($v_{k,s} \neq 0$) are part of the basis, and the omega stringer and the skin form closed cells and, with the periodic condition, the skin itself behaves as a closed contour (Section 2.2.3), so that $\gamma_k = 0$ cannot be satisfied by all distortional modes. Table 1 lists the generalised strains of a mode for a unit amplitude function and its derivatives.

Table 1. Non-zero generalised (mid-surface) strains and curvatures of one mode, multiplied by the amplitude function or its derivatives; terms in brackets vanish under Vlasov's assumptions.

Mode	multiplied by ϕ_k / φ_j / χ_r	multiplied by $\phi_{k,x}$ / $\varphi_{j,x}$ / $\chi_{r,x}$	multiplied by $\phi_{k,xx}$
conventional k	$[\varepsilon^M_{ss} = v_{k,s}]$, $\kappa_{ss} = -w_{k,ss}$	$[\gamma^M_{xs} = \gamma_k]$, $\kappa_{xs} = -2w_{k,s}$	$\varepsilon^M_{xx} = u_k$, $\kappa_{xx} = -w_k$
shear j	$\gamma^M_{xs} = u_{j,s}$	$\varepsilon^M_{xx} = u_j$	–
Mindlin r	$\kappa_{xs} = \xi_{r,s}$, $\kappa_{ss} = \zeta_{r,s}$, $\gamma_{xz} = \xi_r$, $\gamma_{sz} = \zeta_r$	$\kappa_{xx} = \xi_r$, $\kappa_{xs} = \zeta_r$	–

The geometrically non-linear membrane strains needed for the geometric stiffness are

$$\varepsilon_{xx}^{NL} = \tfrac{1}{2}\left(u_{,x}^2 + v_{,x}^2 + w_{,x}^2\right), \qquad \varepsilon_{ss}^{NL} = \tfrac{1}{2}\left(u_{,s}^2 + v_{,s}^2 + w_{,s}^2\right), \qquad \gamma_{xs}^{NL} = u_{,x}u_{,s} + v_{,x}v_{,s} + w_{,x}w_{,s}, \quad (5)$$

of which classical GBT retains $\tfrac{1}{2}\left(v_{,x}^2 + w_{,x}^2\right) = \tfrac{1}{2}\left(v_i v_k + w_i w_k\right)\phi_{i,x}\phi_{k,x}$ [30,31].

2.1.3. Constitutive relations

In every layer p of a wall, the in-plane stresses follow from the reduced stiffness components $\overline{Q}_{ij}^{(p)}$ $(i,j = 1,2,6)$ referred to the local axes (x,s), and the transverse shear stresses from $\overline{\mathbf{Q}}_s^{(p)}$. Integration through the thickness gives the membrane, coupling and bending stiffnesses and the transverse shear stiffness [50–52],

$$\left(A_{ij}, B_{ij}, D_{ij}\right) = \sum_{p=1}^{N_L} \int_{z_p}^{z_{p+1}} \overline{Q}_{ij}^{(p)} \left(1, z, z^2\right) dz, \qquad \mathbf{A}_s = \frac{5}{6}\sum_{p=1}^{N_L} \overline{\mathbf{Q}}_s^{(p)} \left(z_{p+1} - z_p\right), \quad (6)$$

and the stress resultants of a wall are

$$\begin{bmatrix} \mathbf{N} \\ \mathbf{M} \\ \mathbf{Q} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} & \mathbf{0} \\ \mathbf{B} & \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_s \end{bmatrix} \begin{bmatrix} \varepsilon^M \\ \kappa \\ \gamma_z \end{bmatrix}, \quad (7)$$

with $\mathbf{N} = [N_{xx}, N_{ss}, N_{xs}]^\mathsf{T}$, $\mathbf{M} = [M_{xx}, M_{ss}, M_{xs}]^\mathsf{T}$, $\mathbf{Q} = [Q_x, Q_s]^\mathsf{T}$, $\varepsilon^M = [\varepsilon_{xx}^M, \varepsilon_{ss}^M, \gamma_{xs}^M]^\mathsf{T}$, $\kappa = [\kappa_{xx}, \kappa_{ss}, \kappa_{xs}]^\mathsf{T}$ and $\gamma_z = [\gamma_{xz}, \gamma_{sz}]^\mathsf{T}$. All entries of $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{D}$ are retained: extension–bending, extension–shear and bending–twisting coupling, as in [31], and the membrane–bending coupling of unsymmetric laminates [49]. Since the transverse extension is a degree of freedom, N_{ss} is not condensed out.

A stringer foot consists of the skin and the flange, which are tied in the shell model; in GBT they form one wall referred to the skin mid-plane. With $\mathbf{A}_0$, $\mathbf{B}_0$, $\mathbf{D}_0$ of the combined stack about its own mid-plane and z_f the distance between the two mid-planes,

$$\mathbf{A}_f = \mathbf{A}_0, \qquad \mathbf{B}_f = \mathbf{B}_0 + z_f\,\mathbf{A}_0, \qquad \mathbf{D}_f = \mathbf{D}_0 + 2z_f\,\mathbf{B}_0 + z_f^2\,\mathbf{A}_0. \quad (8)$$

In the shell model the stringer mid-surface is tied to the skin with the offset $c = (t_s + t_f)/2$, where t_s and t_f are the thicknesses of skin and stringer. In GBT every web root is a node at $Y = c$, joined to the skin node below it by a link wall of height c, which transmits the in-plane displacements and the rotation like a line of rigid offset links without adding axial or twisting stiffness:

$$\begin{gathered} A_{11}^L = 10^{-3}A_{0,11}, \quad A_{22}^L = k_L A_{0,22}, \quad A_{66}^L = k_L A_{0,66}, \\ D_{11}^L = 10^{-3}D_{0,11}, \quad D_{22}^L = k_L D_{0,22}, \quad D_{66}^L = 10^{-3}D_{0,66}, \quad \mathbf{A}_s^L = k_L \mathbf{A}_{s0}, \end{gathered} \quad (9)$$

with the skin stiffnesses $\mathbf{A}_0$, $\mathbf{D}_0$, $\mathbf{A}_{s0}$, $k_L = 10^3$ and all other entries zero. The results do not depend on k_L, whereas a stiff twisting term D_{66}^L would lock the rotation of the web root (Section 3.6).

2.1.4. Nodal description and deformation mode families

Within every segment p of width b_p between the nodes q and q', with $\eta = (s - S_q)/b_p \in [0,1]$, the profiles u and v vary linearly and are described by their nodal values. The transverse displacement is the cubic Hermite interpolation of the nodal normal displacements w_q, $w_{q'}$ and the nodal rotations θ_q, $\theta_{q'}$ $(\theta = w_{,s})$,

$$\begin{gathered} w(\eta) = H_1(\eta)\,w_q + b_p H_2(\eta)\,\theta_q + H_3(\eta)\,w_{q'} + b_p H_4(\eta)\,\theta_{q'}, \\ H_1 = 1 - 3\eta^2 + 2\eta^3, \quad H_2 = \eta - 2\eta^2 + \eta^3, \quad H_3 = 3\eta^2 - 2\eta^3, \quad H_4 = -\eta^2 + \eta^3. \end{gathered} \quad (10)$$

Equation (10) spans the same space as Schardt's description of the transverse bending by nodal moments [24], which is used in the GBT–Ritz approach [46,49]; the nodal form is used here because it turns the periodic conditions into linear relations between nodal quantities. The rotation is shared by all walls that meet at a node (rigid fold lines). With the in-plane nodal displacements $\mathbf{U} = [\mathbf{U}_1; \dots; \mathbf{U}_N] \in \mathbb{R}^{2N}$, $\mathbf{U}_q = (U_{X,q}, U_{Y,q})$, a segment with unit tangent $\mathbf{t}$ and unit normal $\mathbf{n}$ has $v = \mathbf{t} \cdot \mathbf{U}$ and $w = \mathbf{n} \cdot \mathbf{U}$ at its end nodes, and $\Theta \in \mathbb{R}^N$ collects the nodal rotations. Let $\mathbf{E}_\ell$ be the matrix of segment elongations, $(\mathbf{E}_\ell \mathbf{U})_p = \mathbf{t}_p \cdot (\mathbf{U}_{q'} - \mathbf{U}_q)$. Following the organisation of the cross-section analysis by Bebiano et al. [35] and the shear modes of Gonçalves et al. [36], the conventional modes are generated in the families

- **G (global):** rigid in-plane translations and rotation of the cross-section;
- **D (distortional):** in-plane nodal displacements without segment elongation, $\mathbf{E}_\ell \mathbf{U} = \mathbf{0}$, orthogonal to G and L;
- **L (local):** a unit normal displacement of one node that is not a wall junction, and a pure rotation of one node;
- **T (transverse extension):** the segment elongation patterns, i.e. the columns of $\mathbf{E}_\ell^{\mathsf{T}}$;

and the non-conventional modes in the families

- **S (shear):** the nodal hat functions $h_q(s)$ of the warping, one of which is replaced by the axial extension mode $u_j = 1$;
- **M (Mindlin):** for β_x, a nodal vector Ω_q in the plane of the cross-section with $\xi_r = \mathbf{t} \cdot \Omega_q$, interpolated linearly on every segment (one vector at a node inside a straight wall, two at a fold line); for β_s, a nodal value of ζ_r, interpolated linearly.

Since $\mathbb{R}^{2N} = \ker \mathbf{E}_\ell \oplus \operatorname{range} \mathbf{E}_\ell^{\mathsf{T}}$, the families G, D, L and T span all in-plane nodal displacements, and the rotation modes make the nodal rotations independent; the conventional modes are therefore a complete description of the in-plane deformation of the discretised cross-section, organised in the GBT families. In classical GBT the warping profile u_k of the G and D modes follows from Vlasov's condition $\gamma_k = u_{k,s} + v_k = 0$ [24,53]. Since this condition cannot be met in closed contours, the warping is taken as the stiffness-weighted least-squares solution

$$\min_{\mathbf{u}} \Pi_w(\mathbf{u}) = \sum_p \hat{A}_{66,p}\, b_p \left(\frac{u_{q'} - u_q}{b_p} + v_p(\eta = \tfrac{1}{2}) \right)^2 \quad \text{subject to} \quad \mathbf{c}_{ax}^{\mathsf{T}} \mathbf{u} = 0, \tag{11}$$

where $\mathbf{u}$ collects the nodal warping values, $\hat{A}_{66} = A_{66} - A_{26}^2/A_{22}$, and $c_{ax,q} = \sum_{p \ni q} \frac{1}{2} \hat{A}_{11,p}\, b_p$ with $\hat{A}_{11} = A_{11} - A_{12}^2/A_{22}$, so that the warping carries no axial resultant. Π_w is the membrane shear energy per unit length of a unit $\phi_{k,x}$; for an open contour its minimum is zero and Eq. (11) returns the classical warping, while in closed contours the remaining membrane shear strain γ_k is retained in the strains of Eq. (4) and complemented by the shear modes.

2.2. Periodic boundary conditions in the cross-section analysis

2.2.1. Continuous condition and its modal form

The cell is one period of an infinitely wide panel. Every field must therefore take the same value at $X = -W$ and $X = 0$, for every x:

$$\mathcal{U}_\alpha(X = -W,\, x) - \mathcal{U}_\alpha(X = 0,\, x) = 0, \qquad \alpha = 1, \dots, 6, \qquad 0 \le x \le L, \tag{12}$$

where $\mathcal{U}_\alpha$ are the three translations and three rotations of a shell node. This is the constraint set that a shell model imposes with one equation per pair of side nodes and degree of freedom [5–7]. On the skin side edges (normal Y), the shell degrees of freedom correspond to the GBT quantities of Table 2.

Table 2. Degrees of freedom of a skin side node and the corresponding modal conditions.

Shell DOF	Meaning at a skin side node	Modal condition
$\mathcal{U}_1$	in-plane displacement U_X	$U_{X,s_L} = U_{X,s_R}$ for every conventional mode, Eq. (14)

Shell DOF	Meaning at a skin side node	Modal condition
$\mathcal{U}_2$	deflection U_Y	$U_{Y,s_L} = U_{Y,s_R}$ for every conventional mode, Eq. (14)
$\mathcal{U}_3$	axial displacement u	$u_k(s_L) = u_k(s_R)$, Eq. (15); $u_j(s_L) = u_j(s_R)$, Eq. (16)
$\mathcal{U}_4$	rotation $\phi_x = -w_{,x} + \beta_x$	follows from $\mathcal{U}_2$ for $w_{,x}$; $\xi_r(s_L) = \xi_r(s_R)$, Eq. (16)
$\mathcal{U}_6$	rotation $\theta + \beta_s$	$\theta_{s_L} = \theta_{s_R}$, Eq. (14); $\zeta_r(s_L) = \zeta_r(s_R)$, Eq. (16)
$\mathcal{U}_5$	drilling rotation	not a degree of freedom of the plate theory

Inserting Eqs. (1) and (2) into Eq. (12) and collecting the terms of each amplitude function gives, for example for $\mathcal{U}_3$,

$$[u_k(s_L) - u_k(s_R)]\,\phi_{k,x}(x) + \left[u_j(s_L) - u_j(s_R)\right]\varphi_j(x) = 0 \qquad \forall\, x \in [0, L]. \tag{13}$$

Proposition. Equation (12) holds for arbitrary amplitude functions if, and only if, every deformation mode satisfies it individually, i.e. $\mathbf{U}_k(s_L) = \mathbf{U}_k(s_R)$, $\theta_k(s_L) = \theta_k(s_R)$ and $u_k(s_L) = u_k(s_R)$ for all conventional modes, $u_j(s_L) = u_j(s_R)$ for all shear modes and $\xi_r(s_L) = \xi_r(s_R)$, $\zeta_r(s_L) = \zeta_r(s_R)$ for all Mindlin modes.

Proof. Sufficiency is evident from Eq. (13) and its analogues. For necessity, choose all amplitude functions but one equal to zero: Eq. (13) then requires the bracket of the remaining mode to vanish, since $\phi_{k,x}$ or φ_j is not identically zero. □

The proposition is the basis of the present formulation: the periodic condition is a property of the cross-section analysis, not of the member analysis. A basis that satisfies it mode by mode carries the condition into every combination and every truncation of the modes, and the member analysis (Section 2.3) needs neither penalty terms nor Lagrange multipliers. Conversely, a classical GBT basis of an open contour generally violates it mode by mode, so that the condition would have to be imposed as a constraint between amplitude functions.

2.2.2. Conventional modes: constrained family subspaces

For the conventional modes, the conditions on $\mathcal{U}_1$, $\mathcal{U}_2$ and the rotation part of $\mathcal{U}_6$ are three rows acting on the stacked nodal vector,

$$\mathbf{R}\begin{bmatrix}\mathbf{U}\\ \Theta\end{bmatrix} = \mathbf{0}, \qquad \mathbf{R} = \begin{bmatrix}\mathbf{e}_{X,s_L}^{\mathsf{T}} - \mathbf{e}_{X,s_R}^{\mathsf{T}}\\ \mathbf{e}_{Y,s_L}^{\mathsf{T}} - \mathbf{e}_{Y,s_R}^{\mathsf{T}}\\ \mathbf{e}_{\theta,s_L}^{\mathsf{T}} - \mathbf{e}_{\theta,s_R}^{\mathsf{T}}\end{bmatrix} \in \mathbb{R}^{3\times 3N}, \qquad \operatorname{rank}\mathbf{R} = 3, \tag{14}$$

where $\mathbf{e}_{X,q}$, $\mathbf{e}_{Y,q}$ and $\mathbf{e}_{\theta,q}$ extract $U_{X,q}$, $U_{Y,q}$ and θ_q. The admissible in-plane deformations form the subspace $\ker\mathbf{R}$ of dimension $3N-3$. Let $\mathbf{V}_g \in \mathbb{R}^{3N\times n_g}$ hold the nodal fields $[\mathbf{U};\Theta]$ of family $g \in \{G, D, L, T\}$ of the unconstrained cell, with $\operatorname{span}[\mathbf{V}_G\ \mathbf{V}_D\ \mathbf{V}_L\ \mathbf{V}_T] = \mathbb{R}^{3N}$. The periodic modes are constructed family by family, so that every mode keeps the mechanical character of its family:

1. *Unaffected modes.* The columns of $\mathbf{V}_g$ with $\mathbf{R}\mathbf{v} = \mathbf{0}$ (e.g. the local modes of nodes away from the side edges) are kept unchanged.
2. *Constrained combinations within a family.* The remaining columns $\mathbf{V}'_g$ are replaced by $\mathbf{V}'_g\mathbf{Z}_g$, with $\mathbf{Z}_g$ a basis of $\ker(\mathbf{R}\mathbf{V}'_g)$. By construction, $\operatorname{span}(\mathbf{V}_g\mathbf{Z}_g) = \operatorname{span}(\mathbf{V}_g) \cap \ker\mathbf{R}$ is the largest periodic subspace of family g.
3. *Combinations of several families.* The intersection of $\ker\mathbf{R}$ with the sum of the family spaces may contain vectors that do not lie in the sum of the intersections of step 2. Such vectors, $\mathbf{P} = (\mathbf{I} - \Pi_B)\ker\mathbf{R}$ with the orthogonal projector Π_B onto the span of the modes of steps 1 and 2, are added to family T if they elongate a segment ($\mathbf{E}_\ell\mathbf{U} \neq \mathbf{0}$) and to family D otherwise.
4. *Normalisation.* Every mode is scaled to a unit maximum nodal displacement or rotation.

The resulting modal matrix $\mathbf{V} = [\mathbf{V}_G\ \mathbf{V}_D\ \mathbf{V}_L\ \mathbf{V}_T]$ satisfies $\mathbf{RV} = \mathbf{0}$ and $\operatorname{span}\mathbf{V} = \ker\mathbf{R}$, so that no admissible deformation is lost and none is spurious; the number of conventional modes is $n_C = 3N - 3$. Two consequences illustrate the mechanics of the cell. The rigid rotation about the x-axis moves the side nodes by $U_{Y,s_L} - U_{Y,s_R} = -W\omega$ and is not admissible: a torsional rigid-body motion of one cell is incompatible with an infinitely wide panel, so that family G contains only the two translations. The lateral translation of the side nodes and their rotation, on the other hand, are admissible if they are identical at both sides; these combinations of local and global fields enter family D through step 3 and yield 13 distortional modes.

2.2.3. Warping of the conventional modes: the periodic cell as a closed contour

For the warping, $\mathcal{U}_3$ requires $u_k(s_L) = u_k(s_R)$. In terms of the warping problem, the side edges of the skin are thereby joined: the periodic cell becomes a contour that is closed through its side edges, like the rectangular hollow sections and other closed cross-sections of GBT [24,39]. The periodic condition is added to Eq. (11) as a second constraint. With the segment difference matrix $\mathbf{H}$, $(\mathbf{Hu})_p = u_{q'} - u_q$, the weight matrix $\mathbf{W} = \operatorname{diag}(\widehat{A}_{66,p}/b_p)$, the vector $\mathbf{g}$ of the values $b_p\, v_p(\eta = 1/2)$, and the constraint matrix $\mathbf{C}_u = \left[\mathbf{c}_{ax}^{\mathsf{T}};\ \mathbf{e}_{s_L}^{\mathsf{T}} - \mathbf{e}_{s_R}^{\mathsf{T}}\right]$, the functional of Eq. (11) reads $\Pi_w = (\mathbf{Hu} + \mathbf{g})^{\mathsf{T}}\mathbf{W}(\mathbf{Hu} + \mathbf{g})$, and the stationarity of the Lagrangian $\mathcal{L} = \frac{1}{2}\Pi_w + \mu^{\mathsf{T}}\mathbf{C}_u\mathbf{u}$ gives the Karush–Kuhn–Tucker system

$$\begin{bmatrix} \mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{H} & \mathbf{C}_u^{\mathsf{T}} \\ \mathbf{C}_u & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mu \end{bmatrix} = \begin{bmatrix} -\mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{g} \\ \mathbf{0} \end{bmatrix}. \tag{15}$$

$\mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{H}$ is the stiffness matrix of a chain of shear springs; its null space is the uniform warping, which is removed by $\mathbf{c}_{ax}$ (with $\mathbf{c}_{ax}^{\mathsf{T}}\mathbf{1} > 0$), so that Eq. (15) has a unique solution. For a mode with non-zero circulation $\oint v\, ds$ around a closed contour (here including the contour closed through the side edges), the residual γ_k does not vanish; it is the membrane shear flow of a closed section, carried by the term $A_{66}\gamma_k\gamma_i$ of the modal tensors (Section 2.3).

2.2.4. Shear and Mindlin modes: merging of the side-edge functions

The shear and Mindlin modes are nodal, so that $\mathcal{U}_3$, $\mathcal{U}_4$ and $\mathcal{U}_6$ are satisfied by a change of basis. With the matrix $\mathbf{S}$ whose columns are the nodal functions of the unconstrained cell, the periodic basis is $\mathbf{SP}_m$ with the merge matrix $\mathbf{P}_m$, which adds the column of s_L to that of s_R and removes it:

$$h_{s_R} \leftarrow h_{s_R} + h_{s_L}, \qquad \Omega_{s_R} \leftarrow \Omega_{s_R} + \Omega_{s_L}, \qquad \zeta_{s_R} \leftarrow \zeta_{s_R} + \zeta_{s_L}, \qquad h_{s_L},\ \Omega_{s_L},\ \zeta_{s_L} \text{ removed}. \tag{16}$$

A combination $\mathbf{Sa}$ of the unconstrained functions satisfies $a_{s_L} = a_{s_R}$ if, and only if, it lies in $\operatorname{span}(\mathbf{SP}_m)$, so that Eq. (16) spans exactly the periodic subspace. Since the side nodes lie in the straight skin, their tangent spaces coincide and the vectors Ω_{s_L} and Ω_{s_R} can be paired. The axial extension mode is uniform and periodic by construction. The number of shear modes, including the axial mode, is $n_S = N - 1$, and the number of Mindlin modes is $n_M = \sum_q m_q - m_{s_L} + N - 1$, with m_q the number of vectors Ω_q at node q.

2.2.5. Properties of the periodic mode basis

The mode numbers are listed in Table 3. The periodic mode basis has the following properties. (i) *Exactness:* by the proposition of Section 2.2.1, every combination and every truncation of the modes satisfies Eq. (12) identically. (ii) *Completeness:* the modes span exactly the periodic subspace of the nodal description, so that the full basis is equivalent to imposing Eq. (12) node by node. (iii) *Mechanical meaning:* every mode belongs to one of the GBT families, so that the buckling mode can be decomposed into global, distortional, local, transverse-extension, shear and Mindlin contributions. (iv) *Relation to Bloch–Floquet theory:* for the infinite panel with the pitch b_s, the buckling modes are Bloch waves, $\mathbf{u}(X + b_s, x) = e^{\mathrm{i}\kappa b_s}\,\mathbf{u}(X, x)$ [12]; a periodic cell of width $2b_s$ admits exactly the two real cases $\kappa b_s = 0$ and $\kappa b_s = \pi$, i.e. modes that repeat in every bay or alternate in sign from bay to bay.

Table 3. Number of deformation modes per family of the periodic cell (the numbers do not depend on the lay-up).

b_s [mm]	N	G	D	L	T	S (incl. axial)	M	total
150	92	2	13	166	92	91	205	569
285.5	78	2	13	138	78	77	174	482

The periodic condition cannot be added to a truncated set of classical modes without losing these properties. Classical GBT generates the conventional modes from the natural nodes of the open contour and orders them hierarchically; the periodic condition couples the two ends of the contour, removes the rigid rotation, closes the contour for the warping and therefore changes the modes themselves. Imposed afterwards on a truncated classical set, the three rows of Eq. (14) remove three degrees of freedom from a small space and leave too few admissible modes. In the present formulation the constraint is applied before any truncation, and a hierarchical ordering can be introduced afterwards within the periodic subspace. Following classical GBT, the conventional modes are ordered by the generalised eigenvalue problem $\mathbf{B}\,\psi = \Lambda\,(k_r^2\mathbf{D}^I + k_r^4\mathbf{C})\,\psi$ with the modal tensors of Section 2.3 at a reference wavenumber k_r; small Λ indicates a preference for long waves. The first eight such modes of the periodic cell are shown in Figure 2; the two side nodes move identically in every mode, including the lateral rigid translation (mode 2), while the rigid rotation is absent.

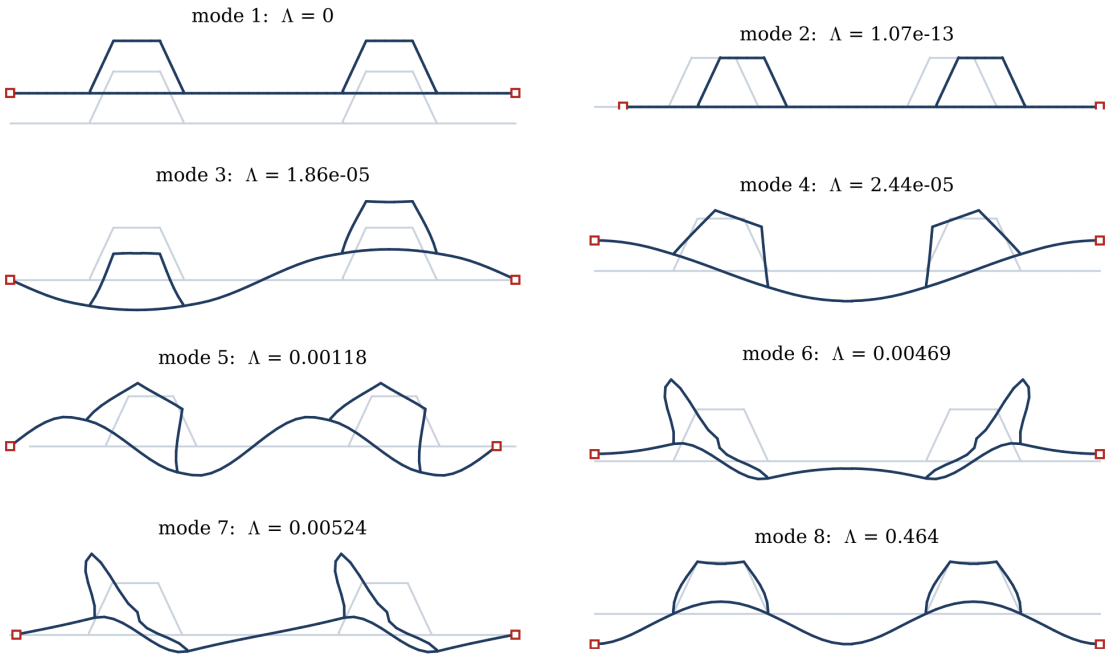


Figure 2. First eight hierarchical conventional modes of the periodic cell ($b_s = 150$ mm, Cross-ply lay-up, $k_r = 2\pi/b_s$; in-plane deformation, grey: undeformed). The side nodes s_L and s_R (squares) move identically in every mode.

2.3. Member analysis

2.3.1. Strain energy variation and modal tensors

With Eqs. (4) and (7), the variation of the strain energy of the cell is, with the stresses $\sigma = [\sigma_{xx}, \sigma_{ss}, \sigma_{xs}, \sigma_{xz}, \sigma_{sz}]^\mathsf{T}$ and the strains $\varepsilon = [\varepsilon_{xx}, \varepsilon_{ss}, \gamma_{xs}, \gamma_{xz}, \gamma_{sz}]^\mathsf{T}$,

$$\delta U = \int_L \int_b \int_t \sigma^\mathsf{T} \delta\varepsilon \, dz \, ds \, dx = \int_L \int_b \left(\mathbf{N}^\mathsf{T} \delta\varepsilon^M + \mathbf{M}^\mathsf{T} \delta\kappa + \mathbf{Q}^\mathsf{T} \delta\gamma_z \right) ds \, dx, \tag{17}$$

Substituting the modal strains of Table 1 and integrating over the cross-section gives, in the form of Silvestre and Camotim [31],

$$\begin{aligned}\delta U = \int_L \Big(& C_{ik}\phi_{k,xx}\delta\phi_{i,xx} + H^I_{ik}\phi_{k,x}\delta\phi_{i,xx} + H^I_{ki}\phi_{k,xx}\delta\phi_{i,x} + D^I_{ik}\phi_{k,x}\delta\phi_{i,x} \\ & + D^{II}_{ik}\phi_k\delta\phi_{i,xx} + D^{II}_{ki}\phi_{k,xx}\delta\phi_i + F^I_{ik}\phi_{k,x}\delta\phi_i + F^I_{ki}\phi_k\delta\phi_{i,x} + B_{ik}\phi_k\delta\phi_i \\ & + C_{hj}\varphi_{j,x}\delta\varphi_{h,x} + H^I_{hj}\varphi_j\delta\varphi_{h,x} + H^I_{jh}\varphi_{j,x}\delta\varphi_h + D^I_{hj}\varphi_j\delta\varphi_h \\ & + C_{ij}\varphi_{j,x}\delta\phi_{i,xx} + H^I_{ij}\varphi_j\delta\phi_{i,xx} + H^{II}_{ji}\varphi_{j,x}\delta\phi_{i,x} \\ & + D^I_{ji}\varphi_j\delta\phi_{i,x} + D^{II}_{ji}\varphi_{j,x}\delta\phi_i + F^{II}_{ij}\varphi_j\delta\phi_i \\ & + C_{kh}\phi_{k,xx}\delta\varphi_{h,x} + H^I_{kh}\phi_{k,xx}\delta\varphi_h + H^{II}_{hk}\phi_{k,x}\delta\varphi_{h,x} \\ & + D^I_{hk}\phi_{k,x}\delta\varphi_h + D^{II}_{hk}\phi_k\delta\varphi_{h,x} + F^{II}_{kh}\phi_k\delta\varphi_h \Big) dx + \delta U_M,\end{aligned} \tag{18}$$

where the tensors with exchanged indices in the last line are those of the third line with the roles of conventional and shear modes exchanged (e.g. C_{kh} is C_{ij} with i and j replaced by k and h), and δU_M contains the terms of the Mindlin modes. The letters of the tensors follow [31]: C pairs the axial strains, B the transverse strains and curvatures, D^I the shear strains and twisting curvatures, D^{II} the axial with the transverse quantities, F the transverse with the shear quantities and H the axial with the shear quantities. For the conventional modes,

$$C_{ik} = \int_b \Big[A_{11}\, u_k u_i - B_{11}\,(w_k u_i + w_i u_k) + D_{11}\, w_k w_i \Big] ds, \tag{19}$$

$$B_{ik} = \int_b \Big[A_{22}\, v_{k,s} v_{i,s} - B_{22}\,(v_{k,s} w_{i,ss} + w_{k,ss} v_{i,s}) + D_{22}\, w_{k,ss} w_{i,ss} \Big] ds, \tag{20}$$

$$D^I_{ik} = \int_b \Big[A_{66}\, \gamma_k \gamma_i - 2B_{66}\,(\gamma_k w_{i,s} + w_{k,s}\gamma_i) + 4D_{66}\, w_{k,s} w_{i,s} \Big] ds, \tag{21}$$

$$D^{II}_{ik} = \int_b \Big[A_{12}\, v_{k,s} u_i - B_{12}\,(v_{k,s} w_i + w_{k,ss} u_i) + D_{12}\, w_{k,ss} w_i \Big] ds, \tag{22}$$

$$F^I_{ik} = \int_b \Big[A_{26}\, \gamma_k v_{i,s} - B_{26}\,(\gamma_k w_{i,ss} + 2 w_{k,s} v_{i,s}) + 2D_{26}\, w_{k,s} w_{i,ss} \Big] ds, \tag{23}$$

$$H^I_{ik} = \int_b \Big[A_{16}\, \gamma_k u_i - B_{16}\,(\gamma_k w_i + 2 w_{k,s} u_i) + 2D_{16}\, w_{k,s} w_i \Big] ds; \tag{24}$$

for the shear modes,

$$C_{hj} = \int_b A_{11}\, u_j u_h\, ds, \qquad D^I_{hj} = \int_b A_{66}\, u_{j,s} u_{h,s}\, ds, \qquad H^I_{hj} = \int_b A_{16}\, u_{j,s} u_h\, ds; \tag{25}$$

and for the coupling between conventional and shear modes,

$$\begin{aligned} C_{ij} &= \int_b (A_{11} u_i - B_{11} w_i)\, u_j\, ds, & H^I_{ij} &= \int_b (A_{16} u_i - B_{16} w_i)\, u_{j,s}\, ds, \\ H^{II}_{ji} &= \int_b (A_{16}\gamma_i - 2B_{16} w_{i,s})\, u_j\, ds, & D^I_{ji} &= \int_b (A_{66}\gamma_i - 2B_{66} w_{i,s})\, u_{j,s}\, ds, \\ D^{II}_{ji} &= \int_b (A_{12} v_{i,s} - B_{12} w_{i,ss})\, u_j\, ds, & F^{II}_{ij} &= \int_b (A_{26} v_{i,s} - B_{26} w_{i,ss})\, u_{j,s}\, ds. \end{aligned} \tag{26}$$

For conventional modes that satisfy Vlasov's assumptions ($\gamma_k = 0$, $v_{k,s} = 0$), Eqs. (19)–(26) are exactly the tensors of Silvestre and Camotim (Eqs. (10)–(12) of [31]); the membrane terms with γ_k and $v_{k,s}$ are the additions required by the closed contours and the transverse extension. The Mindlin terms follow in the same way from Table 1; they contain the transverse shear stiffness, e.g. $\int_b A_{s,11}\, \xi_r \xi_{r'}\, ds$ for two β_x modes,

and the stiffnesses **D** and **B** through the curvatures. In the matrix notation of the GBT–Ritz approach [49], the tensors of the conventional modes form the GBT matrix

$$\Xi = \begin{bmatrix} \mathbf{B} & \mathbf{F}^{I} & \mathbf{D}^{II\mathsf{T}} \\ \mathbf{F}^{I\mathsf{T}} & \mathbf{D}^{I} & \mathbf{H}^{I\mathsf{T}} \\ \mathbf{D}^{II} & \mathbf{H}^{I} & \mathbf{C} \end{bmatrix}, \qquad \delta U = \int_L \delta\Phi^{\mathsf{T}} \Xi\, \Phi \, dx, \qquad \Phi = \begin{bmatrix} \phi \\ \phi_{,x} \\ \phi_{,xx} \end{bmatrix}, \tag{27}$$

with the row index i and the column index k in every block, and it is extended by the shear, coupling and Mindlin tensors in the same arrangement for the complete vector of amplitude functions Φ.

2.3.2. Second-order energy and geometric tensors

The pre-buckling state under the reference load is described by the membrane forces $S_{xx}(x,s)$, $S_{ss}(x,s)$ and $S_{xs}(x,s)$ (Section 2.3.3). Their second-order work on the non-linear strains of Eq. (5) is

$$V = \int_L \int_b (S_{xx}\, \varepsilon_{xx}^{NL} + S_{ss}\, \varepsilon_{ss}^{NL} + S_{xs}\, \gamma_{xs}^{NL})\, ds\, dx, \tag{28}$$

and its variation defines the geometric tensors. For the conventional modes the displacement gradients are $u_{,x} = u_k \phi_{k,xx}$, $u_{,s} = u_{k,s}\phi_{k,x}$, $v_{,x} = v_k \phi_{k,x}$, $v_{,s} = v_{k,s}\phi_k$, $w_{,x} = w_k \phi_{k,x}$ and $w_{,s} = w_{k,s}\phi_k$, and

$$\begin{aligned} \delta V = \int_L \Big(& X_{ik}^{U}\, \phi_{k,xx} \delta\phi_{i,xx} + X_{ik}^{VW}\, \phi_{k,x} \delta\phi_{i,x} + X_{ik}^{S}\, \phi_k \delta\phi_i \\ & + X_{ik}^{T}\, \phi_{k,x} \delta\phi_i + X_{ki}^{T}\, \phi_k \delta\phi_{i,x} + X_{ik}^{R}\, \phi_{k,xx} \delta\phi_{i,x} + X_{ki}^{R}\, \phi_{k,x} \delta\phi_{i,xx} \Big) dx + \dots , \\ X_{ik}^{U} = \int_b S_{xx}\, u_i u_k\, ds, \qquad & X_{ik}^{VW} = \int_b \left[S_{xx}\, (v_i v_k + w_i w_k) + S_{ss}\, u_{i,s} u_{k,s} \right] ds, \\ X_{ik}^{S} = \int_b S_{ss}\, (v_{i,s} v_{k,s} + w_{i,s} w_{k,s})\, ds, \qquad & X_{ik}^{T} = \int_b S_{xs}\, (v_{i,s} v_k + w_{i,s} w_k)\, ds, \\ & X_{ik}^{R} = \int_b S_{xs}\, u_{i,s} u_k\, ds, \end{aligned} \tag{29}$$

where the dots stand for the analogous terms of the shear modes ($u_{,x} = u_j \varphi_{j,x}$, $u_{,s} = u_{j,s}\varphi_j$) and of their coupling with the conventional modes; the Mindlin modes do not move the mid-surface and do not contribute. Since the membrane forces vary along x, the geometric tensors are functions of x. For a uniform axial membrane force S_{xx} and $S_{ss} = S_{xs} = 0$, only X_{ik}^{U} and $X_{ik}^{VW} = \int_b S_{xx}(v_i v_k + w_i w_k)\, ds$ remain; these are the geometric matrices $\mathcal{X}^{U}$ and $\mathcal{X}^{VW}$ of the GBT–Ritz approach [46,49], and X_{ik}^{VW} is the geometric tensor X_{pik} of classical GBT [30,31] for the axial resultant.

2.3.3. Loads and pre-buckling state

The cell is loaded at both loaded edges by a compressive line load $n_0(s)$ per unit length of the mid-line, which acts at the distance $e_0(s)$ from the wall reference surface (Section 3). The variation of the potential of the external forces is

$$\delta\Pi_{ext} = -\int_b n_0\, [\delta u_0(0,s) - \delta u_0(L,s)]\, ds, \qquad u_0 = u - e_0\, w_{,x}, \tag{30}$$

$$\begin{aligned} \delta\Pi_{ext} = & -\left(q_k^I\, \delta\phi_{k,x} + q_h^I\, \delta\varphi_h \right)_{x=0} + \left(q_k^I\, \delta\phi_{k,x} + q_h^I\, \delta\varphi_h \right)_{x=L}, \\ q_k^I = \int_b n_0\, (u_k - e_0 w_k)\, ds, \qquad & q_h^I = \int_b n_0\, u_h\, ds, \end{aligned} \tag{31}$$

i.e. the end load enters through the generalised end forces q_k^I and q_h^I associated with the warping of the modes, as the forces q_i^I in [31]. The pre-buckling state is the first-order solution of $\delta U + \delta\Pi_{ext} = 0$ under the end and side conditions of the cell. Its membrane forces

$$[S_{xx},\ S_{ss},\ S_{xs}]^{\mathsf{T}} = \mathbf{A}\,\varepsilon^{M,0} + \mathbf{B}\,\kappa^{0} \tag{32}$$

vary along s and x: the load enters the stringer through the eccentric foot and is shared between skin and stringer by shear lag, the eccentric foot load bends the cell near the ends, and the restrained Poisson expansion produces transverse membrane forces S_{ss}. In classical GBT the pre-buckling state is expressed by the modal resultants W_p^o of an assumed stress distribution [31]; here it is the computed first-order state, which is necessary because the load is introduced eccentrically at the ends of a stiffened cell.

2.3.4. Equilibrium equations and boundary conditions

The buckling problem follows from the stationarity of the total potential energy of the adjacent state, $\delta U + \lambda\,\delta V = 0$, where λ multiplies the reference load. Integrating Eq. (18) by parts and collecting the coefficients of $\delta\phi_i$ and $\delta\varphi_h$ gives the GBT equilibrium equations, written for the terms of the conventional and shear modes:

$$\begin{aligned} \mathcal{K}_{ik}\,\phi_k + \mathcal{K}_{ij}\,\varphi_j + \lambda\left(\mathcal{G}_{ik}\,\phi_k + \mathcal{G}_{ij}\,\varphi_j\right) = 0,\\ \mathcal{K}_{hk}\,\phi_k + \mathcal{K}_{hj}\,\varphi_j + \lambda\left(\mathcal{G}_{hk}\,\phi_k + \mathcal{G}_{hj}\,\varphi_j\right) = 0, \end{aligned} \tag{33}$$

with the stiffness operators

$$\begin{aligned} \mathcal{K}_{ik} &= C_{ik}\frac{d^4}{dx^4} + H_{ik}\frac{d^3}{dx^3} - D_{ik}\frac{d^2}{dx^2} + F_{ik}\frac{d}{dx} + B_{ik},\\ \mathcal{K}_{ij} &= C_{ij}\frac{d^3}{dx^3} + \left(H_{ij}^{I} - H_{ji}^{II}\right)\frac{d^2}{dx^2} - \left(D_{ji}^{I} - D_{ji}^{II}\right)\frac{d}{dx} + F_{ij}^{II},\\ \mathcal{K}_{hk} &= C_{kh}\frac{d^3}{dx^3} + \left(H_{hk}^{II} - H_{kh}^{I}\right)\frac{d^2}{dx^2} - \left(D_{hk}^{I} - D_{hk}^{II}\right)\frac{d}{dx} - F_{kh}^{II},\\ \mathcal{K}_{hj} &= C_{hj}\frac{d^2}{dx^2} + H_{hj}\frac{d}{dx} - D_{hj}^{I}, \end{aligned} \tag{34}$$

where the second equation of Eq. (33) has been multiplied by -1, as in [31], and

$$H_{ik} = H_{ik}^{I} - H_{ki}^{I}, \qquad D_{ik} = D_{ik}^{I} - D_{ik}^{II} - D_{ki}^{II}, \qquad F_{ik} = F_{ik}^{I} - F_{ki}^{I}, \qquad H_{hj} = H_{hj}^{I} - H_{jh}^{I}. \tag{35}$$

For the uniform membrane force of classical GBT, the geometric operator of the conventional modes is $\mathcal{G}_{ik} = X_{ik}^{U}\,d^4/dx^4 - X_{ik}^{VW}\,d^2/dx^2$; in general it follows from Eq. (29) with coefficients that depend on x. The boundary terms of the integration by parts give the boundary conditions at $x = 0$ and $x = L$,

$$\left[\left(W_i^{\sigma} + q_i^{I}\right)\delta\phi_{i,x} + W_i^{\tau}\,\delta\phi_i + \left(W_h^{\sigma} + q_h^{I}\right)\delta\varphi_h\right]_{x=0}^{x=L} = 0, \tag{36}$$

with the generalised normal stress resultants

$$\begin{aligned} W_i^{\sigma} &= C_{ik}\phi_{k,xx} + H_{ik}^{I}\phi_{k,x} + D_{ik}^{II}\phi_k + C_{ij}\varphi_{j,x} + H_{ij}^{I}\varphi_j,\\ W_h^{\sigma} &= C_{hj}\varphi_{j,x} + H_{hj}^{I}\varphi_j + C_{kh}\phi_{k,xx} + H_{hk}^{II}\phi_{k,x} + D_{hk}^{II}\phi_k, \end{aligned} \tag{37}$$

and the generalised shear stress resultant $W_i^{\tau} = -W_{i,x}^{\sigma} + W_i^{xs}$ with $W_i^{xs} = H_{ki}^{I}\phi_{k,xx} + D_{ik}^{I}\phi_{k,x} + F_{ki}^{I}\phi_k + H_{ji}^{II}\varphi_{j,x} + D_{ji}^{I}\varphi_j$ (geometric terms included in the buckling problem). Equations (33)–(37) are the equations of Silvestre and Camotim [31], generalised by the membrane terms of the tensors, the Mindlin modes and the non-uniform pre-buckling state. Two features of the panel cell, however, prevent a closed-form or harmonic solution of Eq. (33): the geometric operators have coefficients that vary along x, and the end conditions of a stiffened cell involve combinations of modes (Section 2.3.5). The equations are therefore solved by the Ritz method.

2.3.5. Ritz solution and end conditions

As in the GBT–Ritz approach [46,49], Φ is written as the product of a matrix Ψ of shape functions and the vector **a** of unknown coefficients, $\Phi = \Psi\,\mathbf{a}$, with every amplitude function approximated as

$$\phi_k(x) = \sum_{b=1}^{n_b} N_b(x)\, a_{kb}, \qquad \varphi_j(x) = \sum_{b=1}^{n_b} N_b(x)\, a_{jb}, \qquad \chi_r(x) = \sum_{b=1}^{n_b} N_b(x)\, a_{rb}. \tag{38}$$

For members, trigonometric series are used [49]. In the panel cell, the pre-buckling state varies strongly near the loaded edges, where the load enters the stringer, and the clamped condition has to be combined with free warping at one end. The shape functions N_b are therefore cubic B-splines on an open, uniform knot vector with spans of length $h_x = \min(15\ \text{mm},\ \max(2\ \text{mm},\ W/40),\ L/10)$, i.e. $h_x = b_s/20$ for all panels studied. With the open knot vector, $\phi(0) = a_1$, $\phi(L) = a_{n_b}$ and $\phi_{,x}(0) = 0 \Leftrightarrow a_2 = a_1$, so that the essential end conditions act directly on the end coefficients, and the natural conditions of Eq. (36) are satisfied in the weak sense. Two end conditions of the loaded edges are considered, as in the shell models:

- **Simply supported (SS):** the out-of-plane displacement of every end edge vanishes, $U_Y = 0$ at every node at $x = 0$ and $x = L$, i.e. $\sum_k U_{Y,qk}\,\phi_k(0) = \sum_k U_{Y,qk}\,\phi_k(L) = 0$ for every node q. These linear relations between the end coefficients of the conventional modes are imposed by a null-space reduction; the in-plane displacement U_X, the warping and the rotations of the end cross-sections remain free. Rigid-body motions are removed by fixing the axial mode and U_X of the side node s_R at $x = 0$.
- **Clamped (CC):** $\phi_k = \phi_{k,x} = 0$ at both ends for every conventional mode (in-plane displacements and the rotations $w_{,x}$ and θ vanish), $\chi_r = 0$ at both ends, and $\varphi_j(0) = 0$, so that $u = 0$ on the whole edge $x = 0$; the edge $x = L$ is free to warp under the uniform load.

The classical simply supported condition of GBT, $\phi_k = \sin(m\pi x/L)$, suppresses all in-plane displacements of the end cross-sections while leaving their warping free; the present SS condition is weaker and suppresses only the out-of-plane displacement U_Y of the end edges, as in the shell model. Inserting Eq. (38) into $\delta U + \lambda\,\delta V = 0$ gives the eigenvalue problem

$$(\mathbf{K} + \lambda\,\mathbf{G})\,\mathbf{a} = \mathbf{0}, \qquad \mathbf{K} = \int_0^L \Psi^{\mathsf{T}}\,\Xi\,\Psi\,dx, \qquad \mathbf{G} = \int_0^L \Psi^{\mathsf{T}}\,\mathcal{X}(x)\,\Psi\,dx, \tag{39}$$

with Ξ the complete matrix of the modal tensors and $\mathcal{X}(x)$ that of the geometric tensors of Eq. (29). The integrals are evaluated with four Gauss points per span. The smallest positive eigenvalue multiplies the reference load of 1 N/mm, so that $N_{cr} = \lambda \times 1$ N/mm, and **a** defines the buckling mode. Since **G** is indefinite (the pre-buckling state contains membrane shear and some transverse tension), the problem is solved for the largest $\mu = 1/\lambda$ of $-\mathbf{G}\mathbf{a} = \mu\mathbf{K}\mathbf{a}$ by the implicitly restarted Lanczos method [54] with a sparse factorisation of **K**. For $b_s = 150$ mm and $L = 225$ mm the model has $569 \times 33 = 18\,777$ coefficients before the end conditions are applied. Two numerical aspects are noted. First, at some intermediate nodes of the inclined webs the two segment tangents coincide up to round-off, and the orthonormal basis of the tangents contains a spurious Mindlin vector normal to the web, with $\xi_r = \mathbf{t}\cdot\Omega \approx 10^{-15}$ (11 such modes for $b_s = 150$ mm and 8 for $b_s = 285.5$ mm); these modes carry neither energy nor displacement gradients and are held. Second, at a cross-section integration point only about 25 of the 569 modes are non-zero, so that the cross-section factors of $\mathcal{X}$ are stored as sparse matrices that are computed once per cross-section.

2.3.6. Classical limit

For the conventional modes alone, walls without coupling ($\mathbf{B} = \mathbf{0}$, $A_{16} = A_{26} = D_{16} = D_{26} = 0$) and a uniform membrane force S_{xx}, the tensors H_{ik} and F_{ik} vanish and Eq. (33) becomes

$$C_{ik}\,\phi_{k,xxxx} - D_{ik}\,\phi_{k,xx} + B_{ik}\,\phi_k + \lambda\left(X^U_{ik}\,\phi_{k,xxxx} - X^{VW}_{ik}\,\phi_{k,xx}\right) = 0, \tag{40}$$

which, without the warping term X^U_{ik}, is the classical second-order GBT equation of Schardt [24] and of Silvestre and Camotim [30]. For simply supported ends and $\phi_k = \hat{\phi}_k \sin(m\pi x/L)$, with $k_m = m\pi/L$,

$$\left[B_{ik} + k_m^2\,D_{ik} + k_m^4\,C_{ik} + \lambda\,(k_m^2\,X^{VW}_{ik} + k_m^4\,X^U_{ik})\right]\hat{\phi}_k = 0, \tag{41}$$

whose smallest eigenvalue over m is the classical GBT signature curve. The periodic condition does not change this structure; it restricts the admissible cross-section modes. Equation (41) was used to verify the implementation (Section 3.1).

2.4. Mode decomposition

The eigenvector **a** defines the buckling mode in terms of the GBT modes. As in the GBT–Ritz approach [49], an energy-based mode decomposition quantifies the contribution of the deformation mechanisms. The modes are grouped into the families $g \in \{G, D, L, T, S, M\}$, and $\mathbf{a}_g$ denotes the vector **a** with all coefficients set to zero except those of family g. The strain energy of the buckling mode, $U = \frac{1}{2}\mathbf{a}^\mathsf{T}\mathbf{K}\mathbf{a}$, is divided into the contributions of the families,

$$U_g = \tfrac{1}{2}\,\mathbf{a}_g^\mathsf{T}\mathbf{K}\,\mathbf{a}, \qquad \rho_g = \frac{U_g}{U}, \qquad \sum_g \rho_g = 1. \tag{42}$$

If the families are orthogonal with respect to **K**, U_g equals the energy of family g alone, $\frac{1}{2}\mathbf{a}_g^\mathsf{T}\mathbf{K}\mathbf{a}_g$, the form used for members [49]. In the panel cell the distortional and local modes are not orthogonal, because the D modes move the wall junctions without rotating the nodes and the rotation modes of family L restore the continuity of the walls. The energies of the families alone then overestimate the coupled families and do not add up to U, whereas Eq. (42) divides the strain energy exactly. For the Unsym cell with simply supported loaded edges, $b_s = 150$ mm and $\beta = 1.5$, the energies of the families alone would assign 41 % to the distortional and 58 % to the local family, whereas Eq. (42) gives 1 % and 97 %.

3. Results and discussion

This section verifies the formulation and discusses the buckling behaviour of omega-stringer-stiffened panels with periodic side edges. The critical loads predicted by the GBT–Ritz formulation are compared with Abaqus shell models with periodic constraint equations for simply supported loaded edges and with an independent spline finite strip model for clamped loaded edges. Particular attention is given to the critical loads, the buckling modes, the contribution of the individual deformation modes and the influence of the laminate coupling. The geometry of the cross-section and the material properties are summarised in Table 4, and the lay-ups in Table 5. They include an isotropic reference, a cross-ply laminate without coupling, two symmetric angle-ply laminates with bending–twisting coupling ($D_{16}, D_{26} \neq 0$; Sym2 and the seven-ply stringer of Sym1 also with $A_{16}, A_{26} \neq 0$), and an unsymmetric laminate with membrane–bending coupling ($\mathbf{B} \neq \mathbf{0}$). Ply angles are measured from the stringer axis x. Two stringer pitches, $b_s = 150$ mm and 285.5 mm, and aspect ratios $0.5 \leq \beta \leq 3$ are considered.

Table 4. Geometry of the cross-section (Figure 1) and material properties.

Parameter	Value
stringer pitch b_s	150 mm, 285.5 mm
web root half-width b_3, web height h, web inclination	27.75 mm, 30 mm, 25° from the vertical
crown half-width $h_t = b_3 - h\tan 25^\circ$	13.76 mm
foot width $b_1 = 0.1\,(b_s - 2b_3)$	9.45 mm, 23.00 mm
skin and stringer thickness t_s, t_f	1.0 mm, 0.875 mm (8 and 7 plies of 0.125 mm)
stringer offset $c = (t_s + t_f)/2$	0.9375 mm
CFRP ply	$E_1 = 150$ GPa, $E_2 = 9$ GPa, $G_{12} = G_{13} = G_{23} = 5.3$ GPa, $\nu_{12} = 0.33$
isotropic material	$E = 200$ GPa, $\nu = 0.3$

Table 5. Lay-ups of skin and stringer (angles in degrees from the x-axis; subscript s: symmetric, subscript 2: repeated twice).

Name	Skin	Stringer	Coupling
Isotropic	–	–	none
Cross-ply	$[0/90/0/90]_s$	$[0/90/0/90/0/90/0]$	none
Sym1	$[45/-45/-45/45]_s$	$[45/-45/45/-45/45/-45/45]$	D_{16}, D_{26} (stringer also A_{16}, A_{26})
Sym2	$[60/-30/-30/60]_s$	$[60/-30/60/-30/60/-30/60]$	$A_{16}, A_{26}, D_{16}, D_{26}$
Unsym	$[0/30/60/90]_2$	$[0/30/60/90/0/30/60]$	**A**, **B**, **D** fully populated

A uniform compressive line load of 1 N/mm acts at both loaded edges on the skin, the webs and the crowns, and 2 N/mm on a foot, which carries skin and flange; the foot load acts at the height of its resultant, $Y = c/2$. The eigenvalue λ is therefore the critical line load N_{cr} in N/mm, the quantity reported by the buckling analysis of the shell model.

The shell models are linear buckling analyses in Abaqus with S4R and S3 elements and a seed of 3.7 mm. The stringer is tied to the skin by *TIE with the true offset of the mid-surfaces, and the loads are applied as shell edge loads. The simply supported loaded edges are modelled by $\mathcal{U}_2 = 0$ on every end edge of skin and stringer. For the periodic side edges, every skin node of the edge $X = -W$ is paired with the node of $X = 0$ at the same x, and one constraint equation $\mathcal{U}_\alpha(-W, x) - \mathcal{U}_\alpha(0, x) = 0$ is written for each pair and each of the six degrees of freedom, as in Eq. (12); at the two end corners the equation for $\mathcal{U}_2$ is omitted because $\mathcal{U}_2 = 0$ is already prescribed there. The rigid-body motions left by periodicity are removed by $\mathcal{U}_1 = \mathcal{U}_3 = 0$ at the corner $X = 0$, $x = 0$. The GBT model imposes the same physical conditions (Table 2). Abaqus results with periodic side edges are available for the simply supported loaded edges.

For the clamped loaded edges, an in-house spline finite strip model (FSM) [10,11] serves as the reference. It discretises the same geometry, laminates, loads and end conditions with flat laminated Mindlin strips of width $b_s/50$ (twice as fine as the GBT segments), linear interpolation across each strip, a mid-strip transverse shear strain against shear locking, and six degrees of freedom per nodal line in global axes. Along x it uses the same cubic B-splines as the GBT model. The web roots are joined to the skin by rigid offset links, the direct analogue of the *TIE of the shell model, and the periodic condition is applied node by node by merging the six degrees of freedom of the two side nodal lines for every spline coefficient, which is literally the constraint set of the shell model. The pre-buckling state and the geometric stiffness are computed as in Section 2.3. The FSM shares the plate theory with GBT but none of the cross-section discretisation: it has no deformation modes, a finer mesh, rigid links instead of link walls, and boundary conditions on nodal degrees of freedom instead of built into modes. Its agreement with GBT is therefore an independent check of the mode construction. With simply supported loaded edges it agrees with Abaqus within 0.59 % on average and 1.37 % at most (Table 6).

3.1. Verification of the formulation

Since the periodic condition is built into the modes, every buckling mode satisfies it identically. For the cross-ply cell with $b_s = 150$ mm, the largest difference between the two side nodes over all conventional modes is 7.9×10^{-15} for the in-plane displacements, 1.1×10^{-16} for the rotation θ and 2.3×10^{-11} for the warping (modes scaled to a unit maximum), and it is zero for the shear and Mindlin modes. In the buckling mode of the Unsym panel with clamped loaded edges, $b_s = 150$ mm and $\beta = 1.5$, the largest differences between the side edges along the whole length are 1.2×10^{-18} (U_X), 6.9×10^{-18} (U_Y), 1.4×10^{-20} (θ) and 1.1×10^{-15} (warping), with modal amplitudes of order 10^{-5} to 10^{-1}, while the side edges themselves move.

The explicit modal tensors of Eqs. (19)–(26) were compared with the cross-section integral of Eq. (17) evaluated numerically by the implementation for the Cross-ply, Sym2 and Unsym cells; all blocks agree to within 2×10^{-15} relative to their largest entry, i.e. to round-off. The classical limit of Section 2.3 was checked for the Cross-ply cell with $b_s = 150$ mm by restricting the model to the conventional modes and imposing a uniform compressive membrane force and the classical end condition $\varphi_i(0) = \varphi_i(L) = 0$. The B-spline Ritz solution then reproduces the minimum of the signature-curve eigenvalue problem of Eq. (41), 33.747 N/mm with three half-waves for $L = 225$ mm, to within 0.001 %; the same value is obtained for $L = 450$ mm with six half-waves.

3.2. Simply supported loaded edges

Figure 3 shows the critical load as a function of the aspect ratio for the two stringer pitches and all lay-ups, together with the Abaqus results, and Table 6 lists all 30 cases that were computed with Abaqus. GBT agrees with Abaqus within 0.57 % on average and 1.25 % at most; the FSM agrees within 0.59 % and 1.37 %, and GBT and FSM differ by 0.22 % on average. The accuracy is the same for the uncoupled and the coupled laminates (mean deviation 0.49–0.73 % for every lay-up), which confirms the treatment of the bending–twisting and membrane–bending coupling in Eqs. (19)–(26).

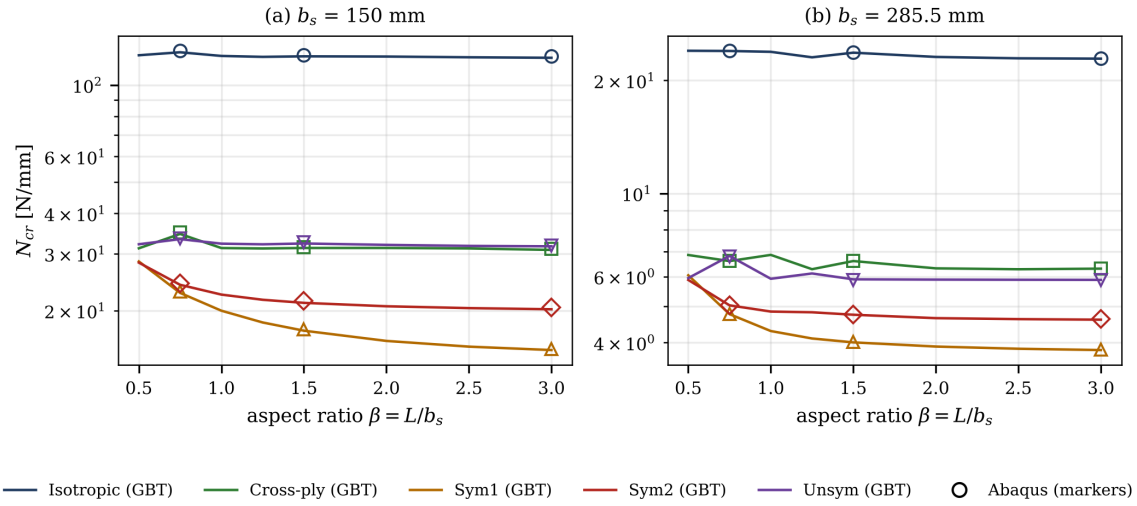


Figure 3. Critical load N_{cr} as a function of the aspect ratio β of the periodic cell with simply supported loaded edges for (a) $b_s = 150$ mm and (b) $b_s = 285.5$ mm: GBT (lines) and Abaqus shell models with periodic constraint equations (markers).

For the isotropic, cross-ply and unsymmetric cells, the critical load varies only little with the aspect ratio, with local maxima where the length of the cell does not accommodate an integer number of buckle half-waves of the preferred length, as known from the garland curves of plates. For $b_s = 150$ mm such a maximum occurs at $\beta = 0.75$, and the Abaqus results follow it (cross-ply: 34.848 N/mm in Abaqus and 34.530 N/mm in GBT at $\beta = 0.75$, against 31.464 and 31.243 N/mm at $\beta = 1.5$). For the angle-ply cells Sym1 and Sym2, the critical load decreases monotonically with β towards the value of the long cell. Their bending–twisting coupling skews the buckles, which therefore cannot adapt to the straight loaded edges by an integer number of half-waves, and the end conditions stiffen short cells.

Table 6. Critical loads N_{cr} [N/mm] with periodic side edges and simply supported loaded edges: Abaqus, GBT and spline finite strip model (FSM), and their deviations e from Abaqus.

Lay-up	b_s [mm]	β	Abaqus	GBT	e_{GBT} [%]	FSM	e_{FSM} [%]
Isotropic	150	0.75	127.650	126.284	−1.07	126.345	−1.02
	150	1.5	123.880	122.674	−0.97	122.617	−1.02
	150	3	122.740	121.412	−1.08	121.395	−1.10
	285.5	0.75	23.979	23.954	−0.10	24.022	+0.18
	285.5	1.5	23.720	23.707	−0.05	23.742	+0.09
	285.5	3	22.873	22.852	−0.09	22.898	+0.11
Cross-ply	150	0.75	34.848	34.530	−0.91	34.606	−0.69
	150	1.5	31.464	31.243	−0.70	31.258	−0.65
	150	3	31.061	30.810	−0.81	30.846	−0.69
	285.5	0.75	6.607	6.595	−0.18	6.618	+0.17
	285.5	1.5	6.609	6.597	−0.18	6.620	+0.17
	285.5	3	6.304	6.296	−0.13	6.310	+0.10
Sym1	150	0.75	22.880	22.675	−0.90	22.583	−1.30
	150	1.5	17.475	17.315	−0.92	17.272	−1.16
	150	3	15.202	15.078	−0.82	15.087	−0.76

Lay-up	b_s [mm]	β	Abaqus	GBT	e_{GBT} [%]	FSM	e_{FSM} [%]
	285.5	0.75	4.760	4.762	+0.04	4.772	+0.25
	285.5	1.5	4.011	4.004	−0.17	4.020	+0.22
	285.5	3	3.832	3.819	−0.34	3.840	+0.21
Sym2	150	0.75	24.291	24.024	−1.10	23.988	−1.25
	150	1.5	21.386	21.118	−1.25	21.144	−1.13
	150	3	20.415	20.174	−1.18	20.229	−0.91
	285.5	0.75	5.033	5.026	−0.14	5.041	+0.16
	285.5	1.5	4.758	4.744	−0.29	4.762	+0.08
	285.5	3	4.620	4.602	−0.39	4.623	+0.06
Unsym	150	0.75	33.489	33.277	−0.63	33.159	−0.99
	150	1.5	32.561	32.260	−0.92	32.115	−1.37
	150	3	31.825	31.616	−0.66	31.511	−0.99
	285.5	0.75	6.775	6.806	+0.46	6.805	+0.44
	285.5	1.5	5.880	5.896	+0.27	5.896	+0.27
	285.5	3	5.864	5.877	+0.22	5.877	+0.22

The deviations depend mainly on the stringer pitch. For $b_s = 285.5$ mm the mean deviation of GBT is 0.20 % (maximum 0.46 %). For $b_s = 150$ mm it is 0.93 % (maximum 1.25 %), and all 15 GBT results and all 15 FSM results lie below Abaqus (GBT −1.25 % to −0.63 %). This offset is therefore common to the two plate models and not caused by the GBT mode basis, and the refinement study of Section 3.6 shows that it is not a discretisation error of GBT either, since finer segments lower the GBT result slightly. Its likely origins are differences between the S4R shell elements and the plate model in the short skin bays of the narrow pitch, and the representation of the tie; it has not been resolved in this study.

3.3. Clamped loaded edges

The critical loads with clamped loaded edges are shown in Figure 4 and listed in Table 7. Clamping raises the critical load of short cells considerably, e.g. by 59 % for the isotropic cell with $b_s = 150$ mm and $\beta = 0.5$, and the curves decrease monotonically with β. For long cells the clamped values approach, and for some lay-ups fall slightly below, the simply supported ones (e.g. 118.13 against 121.41 N/mm for the isotropic cell with $\beta = 3$); the FSM gives the same trend (118.17 against 121.40 N/mm). In the present models the clamped edges also restrain the in-plane displacements of the loaded edges, which changes the pre-buckling state near the ends. GBT and FSM agree within 0.30 % on average and 1.17 % at most over the 30 cases, with the largest deviations for the Sym1 lay-up with $b_s = 285.5$ mm, where the long skin bays and the bending–twisting coupling produce steep gradients of the mode near the clamped ends (Section 3.6). Abaqus results with periodic side edges and clamped loaded edges are not yet available, so that the clamped results are verified by the FSM only.

Table 7. Critical loads N_{cr} [N/mm] with periodic side edges and clamped loaded edges: spline finite strip model (FSM) and GBT, and the deviation e of GBT from the FSM.

Lay-up	b_s [mm]	β	FSM	GBT	e_{GBT} [%]
Isotropic	150	0.75	145.165	145.225	+0.04
	150	1.5	124.834	124.801	−0.03
	150	3	118.166	118.133	−0.03
	285.5	0.75	30.694	30.645	−0.16
	285.5	1.5	24.736	24.675	−0.25
	285.5	3	22.935	22.870	−0.28
Cross-ply	150	0.75	48.899	48.878	−0.04
	150	1.5	36.538	36.503	−0.10
	150	3	32.052	32.027	−0.08
	285.5	0.75	11.243	11.228	−0.13
	285.5	1.5	7.754	7.737	−0.22
	285.5	3	6.716	6.699	−0.25

Lay-up	b_s [mm]	β	FSM	GBT	e_{GBT} [%]
Sym1	150	0.75	23.959	24.047	+0.37
	150	1.5	15.654	15.606	−0.31
	150	3	12.984	12.882	−0.79
	285.5	0.75	5.580	5.564	−0.29
	285.5	1.5	3.858	3.813	−1.17
	285.5	3	3.510	3.471	−1.11
Sym2	150	0.75	30.135	30.252	+0.39
	150	1.5	23.579	23.563	−0.07
	150	3	21.668	21.619	−0.23
	285.5	0.75	6.688	6.683	−0.07
	285.5	1.5	5.030	5.005	−0.50
	285.5	3	4.567	4.530	−0.81
Unsym	150	0.75	35.997	36.129	+0.37
	150	1.5	29.924	30.029	+0.35
	150	3	27.722	27.821	+0.36
	285.5	0.75	7.717	7.718	+0.01
	285.5	1.5	6.014	6.013	−0.02
	285.5	3	5.458	5.455	−0.05

3.4. Mode decomposition and buckling modes

Figure 5 shows the skin deflection of the critical mode of the Unsym cell with $b_s = 150$ mm and $\beta = 1.5$ for both end conditions. In both cases the mode is a local buckling of the skin bays between the stringers; the buckles are skewed by the laminate coupling, and the deflection at the side edges $X = -W$ and $X = 0$ is identical (largest difference 2.2×10^{-16} of the maximum deflection), so that the half-bays at the two sides join into one full bay of the same shape as the central one. With clamped loaded edges the buckles are concentrated away from the loaded edges.

The energy-based mode decomposition of Eq. (42) quantifies these observations for all panels (Table 8). In every case the local family carries 96–101 % of the strain energy of the critical mode, the global family up to 3.5 % and the distortional family between −1.6 % and 1.1 %; the transverse-extension, shear and Mindlin families contribute less than 0.1 %. Contributions slightly above 100 % or below zero are possible in Eq. (42), because U_g contains the interaction energy of family g with the other families; small negative distortional contributions of the angle-ply laminates indicate that the distortional modes interact with the local skin buckles. The global participation is largest for the cross-ply cell with $b_s = 285.5$ mm, i.e. the local buckles of the wide skin bays are accompanied by a small overall bending of the cell. The decomposition does not change significantly with the aspect ratio, i.e. no transition to a distortional or global buckling mode occurs in the range studied.

Table 8. Energy-based participation ρ_g [%] of the mode families in the critical mode, range over $0.5 \le \beta \le 3$ (eight aspect ratios).

Lay-up	Ends	local (L)	distortional (D)	global (G)	T, S, M
Isotropic	SS	97.6 to 98.8	−0.2 to 0.1	1.4 to 2.4	< 0.04
	CC	97.8 to 99.0	−0.3 to 0.0	1.2 to 2.3	< 0.01
Cross-ply	SS	96.1 to 96.8	0.4 to 1.1	2.1 to 3.5	< 0.01
	CC	96.3 to 97.3	0.3 to 1.0	1.8 to 3.3	< 0.01
Sym1	SS	98.8 to 101.2	−1.5 to −0.5	0.2 to 1.7	< 0.05
	CC	99.1 to 101.3	−1.5 to −0.4	0.1 to 1.3	< 0.01
Sym2	SS	98.5 to 101.2	−1.6 to −0.4	0.3 to 1.9	< 0.04
	CC	98.5 to 100.4	−0.7 to −0.2	0.0 to 1.7	< 0.01
Unsym	SS	96.1 to 99.1	0.4 to 1.0	0.0 to 3.4	< 0.09
	CC	97.3 to 98.4	−0.1 to 0.3	1.5 to 2.7	< 0.02

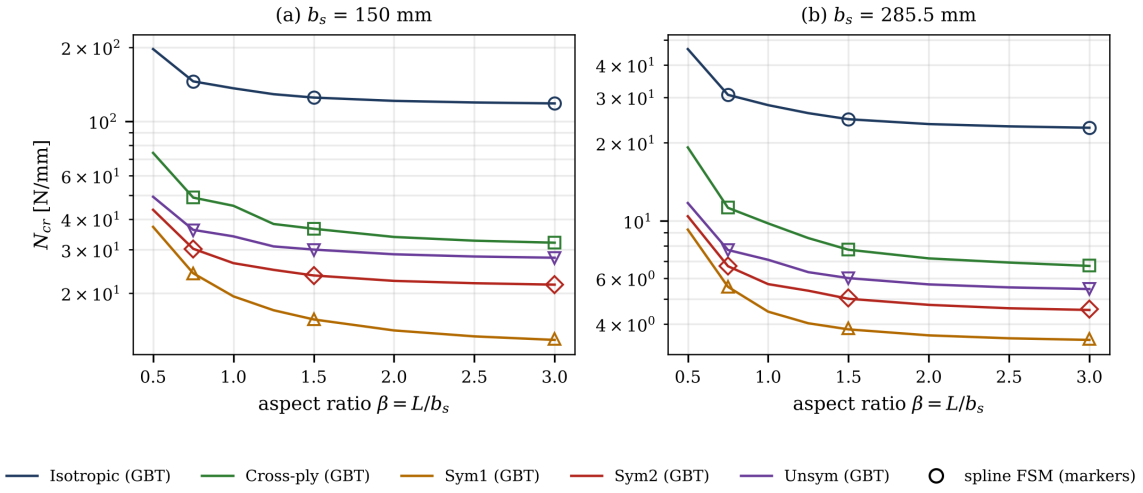


Figure 4. Critical load N_{cr} as a function of the aspect ratio β of the periodic cell with clamped loaded edges for (a) $b_s = 150$ mm and (b) $b_s = 285.5$ mm: GBT (lines) and spline finite strip model (markers).

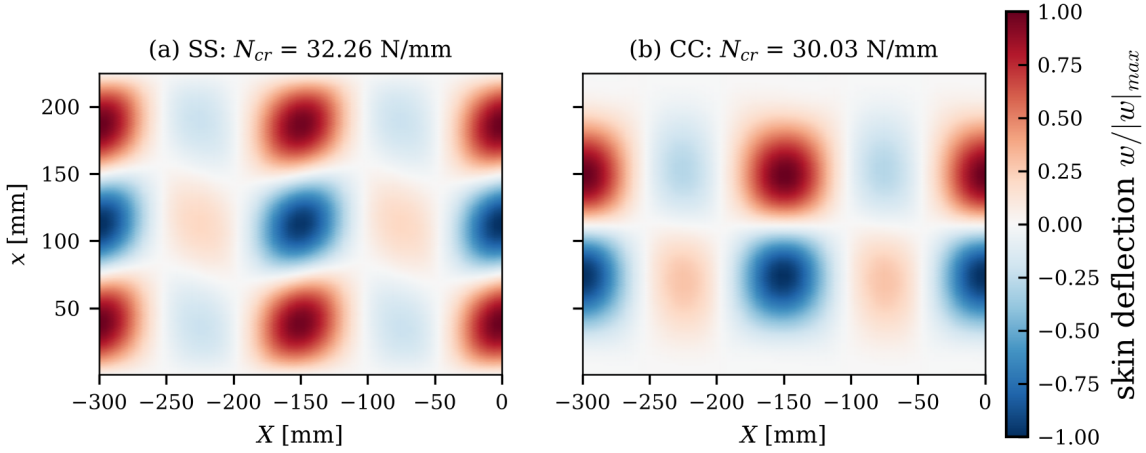


Figure 5. Skin deflection $w/|w|_{max}$ of the critical mode of the Unsym cell with periodic side edges, $b_s = 150$ mm and $\beta = 1.5$: (a) simply supported loaded edges; (b) clamped loaded edges. The stringers lie at $X = -75$ mm and $X = -225$ mm.

3.5. Hierarchical mode bases

The results above use the complete periodic mode basis, which is equivalent to imposing the periodic condition node by node in a folded-plate model with the same nodes. The characteristic feature of GBT is that a small number of hierarchically ordered modes suffices for many problems. This was examined with the hierarchical conventional modes of Section 2.2.5, ordered at $k_r = \pi/b_s$, the pre-buckling state of the complete basis and the end conditions of Section 2.3.5. Table 9 shows the deviation of the critical load when only the first n_C hierarchical conventional modes are kept, together with all shear and Mindlin modes, and when the Mindlin modes are omitted from the complete basis.

Table 9. Deviation [%] of the critical load from the complete basis for truncated mode bases ($\beta = 1.5$): the first n_C hierarchical conventional modes with all shear and Mindlin modes, and all conventional and shear modes without the Mindlin modes.

Lay-up	Ends	b_s [mm]	$n_C = 20$	$n_C = 40$	$n_C = 60$	$n_C = 100$	$n_C = 150$	no Mindlin modes
Isotropic	SS	150	+1.61	+0.53	+0.35	+0.07	+0.03	+0.66
Cross-ply	SS	150	+0.36	+0.15	+0.11	+0.03	+0.01	+0.75
Sym2	SS	150	+3.77	+0.88	+0.56	+0.27	+0.13	+1.67
Unsym	SS	150	+6.03	+3.27	+2.00	+0.83	+0.62	+2.01
Unsym	CC	150	+5.28	+2.72	+1.43	+0.53	+0.48	+1.61
Sym1	CC	285.5	+0.97	+0.16	+0.10	+0.05	+0.02	+0.36

The hierarchical ordering is effective for the conventional modes: for the uncoupled lay-ups, 40 conventional modes reproduce the complete result within 0.53 %, and 100 within 0.07 %. For the coupled lay-ups, more conventional modes are needed (0.2 % to 3.3 % with 40 and 0.05 % to 0.83 % with 100 modes), because the skewed buckles of these laminates combine many local modes. The transverse shear (Mindlin) modes are not negligible: omitting them raises the critical load by 0.36 % to 2.01 %. A small basis in the classical sense, consisting of the first 20 conventional modes, the warping (shear-mode) field that each of them activates at k_r and no Mindlin modes (41 modes), gives +1.45 % for the isotropic cell with $b_s = 285.5$ mm and $\beta = 0.75$ and +1.46 % for the cross-ply cell with $b_s = 150$ mm and $\beta = 1.5$, but +10.36 % and +8.81 % for the Unsym cell with simply supported and clamped loaded edges. Compact GBT bases are therefore adequate for uncoupled laminates, whereas for coupled laminates the complete basis, or a basis that is enriched in the local and Mindlin families, is required; the construction of such bases is left for future work.

3.6. Accuracy and efficiency

The default discretisation (segments of at most $b_s/25$, spline spans $h_x = b_s/20$) was checked by refining each parameter in turn (Table 10). Except for one case, all changes are within 0.50 %, i.e. below the deviation from the reference models. The exception is the Sym1 cell with clamped loaded edges, $b_s = 285.5$ mm and $\beta = 1.5$: halving the spans repeatedly ($h_x/2$, $h_x/4$, $h_x/8$) raises the critical load by +0.26, +0.47 and +0.62 %, i.e. slowly, so that the converged value is estimated to lie about 1 % above the default result. The FSM, which uses the same spline spans, gives 3.858 N/mm for this case, i.e. the largest deviation between GBT and FSM in Table 7 stems mainly from the spline discretisation of GBT near the clamped ends of the wide skin bays with bending–twisting coupling. Finer segments lower the critical load slightly for $b_s = 150$ mm and therefore do not remove the offset against Abaqus noted in Section 3.2. The link walls were checked in the same way: for the isotropic cell with simply supported loaded edges, $b_s = 150$ mm and $\beta = 1.5$, the critical load is 122.667, 122.674 and 122.671 N/mm for $k_L = 10^2$, 10^3 and 10^4, whereas a link wall with the stiff twisting term $D_{66}^L = k_L D_{0,66}$ gives 150.620 N/mm (+22.8 %).

Table 10. Critical load N_{cr} [N/mm] with the default discretisation and with refined spline spans or cross-section segments (periodic side edges); in brackets the change from the default.

Case	Default	Spans $h_x/2$	Segments $b_s/40$
Isotropic, SS, $b_s = 150$, $\beta = 1.5$	122.6740	122.5921 (−0.07 %)	122.4080 (−0.22 %)

Case	Default	Spans $h_x/2$	Segments $b_s/40$
Sym2, SS, $b_s = 150$, $\beta = 3$	20.1744	20.1776 (+0.02 %)	20.1476 (−0.13 %)
Unsym, CC, $b_s = 150$, $\beta = 1.5$	30.0295	30.0275 (−0.01 %)	29.8795 (−0.50 %)
Sym1, CC, $b_s = 285.5$, $\beta = 3$	3.4715	3.4780 (+0.19 %)	3.4698 (−0.05 %)
Sym1, CC, $b_s = 285.5$, $\beta = 1.5$	3.8130	3.8230 (+0.26 %)	3.8100 (−0.08 %)

The model sizes and computing times are compared in Table 11. With the full basis of 482–569 modes, a GBT analysis, including the cross-section analysis, the pre-buckling solution and the eigenvalue problem, takes 4.6 s on average for the ten cells with $b_s = 150$ mm and $\beta = 1.5$ on two cores of a standard workstation; over all 160 cases of Figures 3 and 4, computed two at a time on the same two cores, it ranged from 2.3 to 23.8 s. The GBT model has about half the unknowns of the FSM and less than half the degrees of freedom of the shell model, but with the full basis it is not faster than the FSM (2.5 s): the global and distortional modes extend over the whole cross-section and couple with all other modes, so that its matrices are less sparse than those of the nodal FSM. The cross-section analysis itself takes about 0.1 s and is reused for all lengths of a parametric study. The efficiency of the GBT formulation therefore lies in its modal description, which provides the mode decomposition and is the basis for reduced mode sets, rather than in the solution time of the full basis.

Table 11. Model size and computing time for $b_s = 150$ mm and $L = 225$ mm.

Model	Unknowns	Computing time per case
Abaqus shell model (S4R/S3, seed 3.7 mm)	approx. 5×10^4 DOFs (estimated from the seed)	[to be added by the authors]
spline finite strip model	37 170 (177 nodal lines × 6 × 35 splines)	2.5 s on average
GBT–Ritz (present)	18 777 (569 modes × 33 splines)	4.6 s on average

The following limitations should be noted. First, GBT and the FSM share the plate assumptions: laminated Reissner–Mindlin walls with a shear correction factor of 5/6, an offset tie represented by stiff link walls or rigid links, and linear buckling about a first-order pre-buckling state. An error common to both, such as the −1 % offset for $b_s = 150$ mm, cannot be detected by comparing them. Second, the periodic condition of Eq. (12), like the constraint equations of the shell model, keeps the width of the cell constant. The Poisson expansion of the cell under the axial load is therefore restrained, and transverse membrane forces S_{ss} arise in the pre-buckling state; for the isotropic cell with $b_s = 150$ mm and $\beta = 3$, omitting S_{ss} from $\mathbf{G}$ would raise the critical load by 23 %. The condition thus represents a panel whose lateral expansion is restrained by the surrounding structure. A periodic condition with free lateral expansion, in which the two side edges differ by a common jump in U_X, requires one additional generalised coordinate and is left for future work. Third, a cell of width $2b_s$ admits the Bloch wavenumbers $\kappa b_s = 0$ and π only; buckling modes with a longer period across the stringers require a wider cell. Fourth, compact hierarchical bases are adequate for the uncoupled laminates only (Section 3.5); for the coupled laminates, enriched bases remain to be developed.

4. Conclusions

This study presented a shear-deformable GBT–Ritz formulation for the buckling analysis of omega-stringer-stiffened composite panels in which the periodic boundary conditions of the panel cell are built into the cross-section analysis. The conventional deformation modes are obtained by projecting out the constraint rows of the periodic condition family by family, the warping of the global and distortional modes follows from a constrained Vlasov least-squares problem, and the shear and Mindlin modes are obtained by merging the nodal functions of the two side edges. Every mode, and therefore every combination and truncation of the modes, satisfies the periodic condition to machine precision, so that no penalty terms or Lagrange multipliers are needed in the member analysis. The modal tensors, the equilibrium equations and the boundary

conditions were derived in the notation of the shear-deformable GBT of Silvestre and Camotim, extended by the membrane terms that the closed contours of the cell require, the transverse shear (Mindlin) modes and a computed, non-uniform pre-buckling state; for uncoupled walls under uniform compression the formulation reduces to the classical GBT buckling equations, which was confirmed numerically to 0.001 %. For 30 panels with simply supported loaded edges, the critical loads agree with Abaqus shell models with periodic constraint equations within 0.57 % on average and 1.25 % at most, and for 30 panels with clamped loaded edges they agree with an independent spline finite strip model within 0.30 % on average and 1.17 % at most. The accuracy is the same for cross-ply, angle-ply and unsymmetric laminates, which confirms the consistent treatment of the bending–twisting and membrane–bending coupling.

Beyond accuracy, the formulation provides insight into the buckling mechanisms of the cell. The energy-based mode decomposition identifies the critical modes of all panels studied as local skin buckling, with global and distortional contributions below 4 %, and the variation of the critical load with the aspect ratio of the cell follows directly from the same model at a computing time of a few seconds per case. Since the periodic condition is a property of the modes, hierarchical GBT bases can be formed within the periodic subspace: for uncoupled laminates a few tens of conventional modes reproduce the complete result within about 1 %, whereas the coupled laminates need about 100 conventional modes together with the transverse shear modes, which contribute up to about 2 % to the critical load. Future work will address the confirmation of the clamped cases with shell finite element models, periodic conditions with free lateral expansion of the cell, reduced hierarchical mode bases for parametric studies and surrogate models, and the comparison with the closed-form periodic solutions of El Yaakoubi-Mesbah and Mittelstedt [17].

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References

1. Zhang, G.; Hu, Y.; Yan, B.; Tong, M.; Wang, F. Buckling and post-buckling analysis of composite stiffened panels: A ten-year review (2014–2023). *Thin-Walled Structures* **2024**, *205*, 112525.
2. Mittelstedt, C. *Mechanics of Composite Structures*; Springer: Berlin, Heidelberg, Germany, **2025**. https://doi.org/10.1007/978-3-662-70788-3
3. Mittelstedt, C.; Beerhorst, M. Closed-form buckling analysis of compressively loaded composite plates braced by omega-stringers. *Composite Structures* **2009**, *88*, 424–435. https://doi.org/10.1016/j.compstruct.2008.05.021
4. Schilling, J.C.; Mittelstedt, C. Local buckling analysis of omega-stringer-stiffened composite panels using a new closed-form analytical approximate solution. *Thin-Walled Structures* **2020**, *147*, 106534. https://doi.org/10.1016/j.tws.2019.106534
5. Mittelstedt, C. Closed-form analysis of the buckling loads of uniaxially loaded blade-stringer-stiffened composite plates considering periodic boundary conditions. *Thin-Walled Structures* **2007**, *45*, 371–382. https://doi.org/10.1016/j.tws.2007.04.005
6. Mittelstedt, C. Explicit local buckling analysis of stiffened composite plates accounting for periodic boundary conditions and stiffener–plate interaction. *Composite Structures* **2009**, *91*, 249–265. https://doi.org/10.1016/j.compstruct.2009.04.021
7. Schilling, J.C.; Mittelstedt, C. A simple closed-form analytical model for the column buckling of omega-stringer-stiffened panels with periodic boundary conditions. *Results in Engineering* **2020**, *6*, 100120. https://doi.org/10.1016/j.rineng.2020.100120
8. Wittrick, W.H.; Williams, F.W. Buckling and vibration of anisotropic or isotropic plate assemblies under combined loadings. *International Journal of Mechanical Sciences* **1974**, *16*, 209–239. https://doi.org/10.1016/0020-7403(74)90069-1
9. Kennedy, D.; Williams, F.W.; Anderson, M.S. Buckling and vibration analysis of laminated panels using VICONOPT. *Journal of Aerospace Engineering* **1994**, *7*, 245–262. https://doi.org/10.1061/(ASCE)0893-1321(1994)7:3(245)

10. Cheung, Y.K. *Finite Strip Method in Structural Analysis*; Pergamon Press: Oxford, UK, **1976**.
11. Lau, S.C.W.; Hancock, G.J. Buckling of thin flat-walled structures by a spline finite strip method. *Thin-Walled Structures* **1986**, *4*, 269–294. https://doi.org/10.1016/0263-8231(86)90034-0
12. Mead, D.J. Wave propagation in continuous periodic structures: Research contributions from Southampton, 1964–1995. *Journal of Sound and Vibration* **1996**, *190*, 495–524. https://doi.org/10.1006/jsvi.1996.0076
13. Mittelstedt, C.; Schröder, K.-U. Local postbuckling of hat-stringer-stiffened composite laminated plates under transverse compression. *Composite Structures* **2010**, *92*, 2830–2844. https://doi.org/10.1016/j.compstruct.2010.04
14. Schilling, J.C.; Mittelstedt, C. Shear-buckling of omega-stringer-stiffened panels. *International Journal of Mechanical Sciences* **2021**, *207*, 106646. https://doi.org/10.1016/j.ijmecsci.2021.106646
15. Schilling, J.C.; Atamann, D.; Voges, J.; Mittelstedt, C. Local buckling of omega-stringer-stiffened composite panels under compression–shear interaction. *Thin-Walled Structures* **2022**, *180*, 109838. https://doi.org/10.1016/j.tws.2022.109838
16. Schilling, J.C.; Mittelstedt, C. Local postbuckling of omega-stringer-stiffened composite panels. *Thin-Walled Structures* **2022**, *181*, 110027. https://doi.org/10.1016/j.tws.2022.110027
17. El Yaakoubi-Mesbah, C.; Mittelstedt, C. Closed-form analytical solution for local buckling of omega-stringer-stiffened composite panels under compression. *Composite Structures* **2025**, *353*, 118716. https://doi.org/10.1016/j.compstruct.2024.118716
18. El Yaakoubi-Mesbah, C.; Mittelstedt, C. Computational model for local buckling of compressively loaded omega-stringer-stiffened panels. In Proceedings of the 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics (WCCM-PANACM 2024), Vancouver, Canada, **2024**. https://doi.org/10.23967/wccm.2024.059
19. El Yaakoubi-Mesbah, C.; Mittelstedt, C. Closed-form analytical solution for the postbuckling analysis of omega-stringer-stiffened panels considering stringer feet effects. *Composite Structures* **2026**, *385*, 120209. https://doi.org/10.1016/j.compstruct.2026.120209
20. Mittelstedt, C. Buckling of Stiffened Laminates. In *Mechanics of Composite Structures*; Springer: Berlin, Heidelberg, Germany, **2025**. https://doi.org/10.1007/978-3-662-70788-3
21. Ewing, M.S.; Hinger, R.J.; Leissa, A.W. On the validity of the reduced bending stiffness method for laminated composite plate analysis. *Composite Structures* **1988**, *9*, 301–317. https://doi.org/10.1016/0263-8223(88)90050-5
22. Elsayed, A.M.; Kharghani, N.; El Yaakoubi-Mesbah, C.; Mittelstedt, C. A hybrid physics-informed neural network framework for local buckling prediction of stiffened composite panels with epistemic uncertainty quantification. *Composite Structures* **2026**, *399*, 120959. https://doi.org/10.1016/j.compstruct.2026.120959
23. Schardt, R. *Verallgemeinerte Technische Biegetheorie: Lineare Probleme*; Springer: Berlin, Heidelberg, Germany, **1989**.
24. Schardt, R. Generalized beam theory—An adequate method for coupled stability problems. *Thin-Walled Structures* **1994**, *19*, 161–180.
25. Davies, J.M.; Leach, P. First-order generalised beam theory. *Journal of Constructional Steel Research* **1994**, *31*, 187–220. https://doi.org/10.1016/0143-974X(94)90010-8
26. Camotim, D.; Basaglia, C.; Silvestre, N. GBT buckling analysis of thin-walled steel frames: A state-of-the-art report. *Thin-Walled Structures* **2010**, *48*, 726–743. https://doi.org/10.1016/j.tws.2009.12.003
27. Mittelstedt, C. Buckling and post-buckling of thin-walled composite laminated beams—A review of engineering analysis methods. *Applied Mechanics Reviews* **2020**, *72*, 020802.
28. Mittelstedt, C. Generalized beam theory for the analysis of thin-walled structures—A state-of-the-art survey. *Thin-Walled Structures* **2024**, *200*, 111849.
29. Silvestre, N.; Camotim, D. First-order generalised beam theory for arbitrary orthotropic materials. *Thin-Walled Structures* **2002**, *40*, 755–789. https://doi.org/10.1016/S0263-8231(02)00025-3
30. Silvestre, N.; Camotim, D. Second-order generalised beam theory for arbitrary orthotropic materials. *Thin-Walled Structures* **2002**, *40*, 791–820. https://doi.org/10.1016/S0263-8231(02)00026-5
31. Silvestre, N.; Camotim, D. Shear deformable generalized beam theory for the analysis of thin-walled composite members. *Journal of Engineering Mechanics* **2013**, *139*, 1010–1024.
32. Silvestre, N.; Camotim, D. Generalized beam theory to analyze the vibration of open-section thin-walled composite members. *Journal of Engineering Mechanics* **2013**, *139*, 992–1009.
33. Dinis, P.B.; Camotim, D.; Silvestre, N. GBT formulation to analyse the buckling behaviour of thin-walled members with arbitrarily 'branched' open cross-sections. *Thin-Walled Structures* **2006**, *44*, 20–38.
34. Gonçalves, R.; Ritto-Corrêa, M.; Camotim, D. A new approach to the calculation of cross-section deformation modes in the framework of generalized beam theory. *Computational Mechanics* **2010**, *46*, 759–781. https://doi.org/10.1007/s00466-010-0512-2
35. Bebiano, R.; Gonçalves, R.; Camotim, D. A cross-section analysis procedure to rationalise and automate the performance of GBT-based structural analyses. *Thin-Walled Structures* **2015**, *92*, 29–47. https://doi.org/10.1016/j.tws.2015.02.017

36. Gonçalves, R.; Bebiano, R.; Camotim, D. On the shear deformation modes in the framework of generalized beam theory. *Thin-Walled Structures* **2014**, *84*, 325–334. https://doi.org/10.1016/j.tws.2014.07.012
37. Bebiano, R.; Camotim, D.; Gonçalves, R. GBTul 2.0—A second-generation code for the GBT-based buckling and vibration analysis of thin-walled members. *Thin-Walled Structures* **2018**, *124*, 235–257. https://doi.org/10.1016/j.tws.2017.12.002
38. Silvestre, N. Generalised beam theory to analyse the buckling behaviour of circular cylindrical shells and tubes. *Thin-Walled Structures* **2007**, *45*, 185–198. https://doi.org/10.1016/j.tws.2007.02.001
39. Gonçalves, R.; Camotim, D. Elastic buckling of uniformly compressed thin-walled regular polygonal tubes. *Thin-Walled Structures* **2013**, *71*, 35–45. https://doi.org/10.1016/j.tws.2013.04.016
40. Silva, N.M.F.; Silvestre, N.; Camotim, D. GBT formulation to analyse the buckling behaviour of FRP composite open-section thin-walled columns. *Composite Structures* **2010**, *93*, 79–92. https://doi.org/10.1016/j.compstruct.2010.06.013
41. Silva, N.M.F.; Camotim, D.; Silvestre, N.; Degenhardt, R. On the use of generalized beam theory to assess the buckling and postbuckling behavior of laminated CFRP cylindrical stiffened panels. *International Journal of Structural Stability and Dynamics* **2010**, *10*, 737–760. https://doi.org/10.1142/S0219455410003713
42. Taig, G.; Ranzi, G. Generalised beam theory (GBT) for stiffened sections. *International Journal of Steel Structures* **2014**, *14*, 381–397. https://doi.org/10.1007/s13296-014-2017-x
43. Mittelstedt, C. The Ritz Method for Buckling Analysis of Shear-Deformable Laminates. In *Mechanics of Composite Structures*; Springer: Berlin, Heidelberg, Germany, **2025**; pp. 511–563.
44. Vescovini, R.; Dozio, L.; D'Ottavio, M.; Polit, O. On the application of the Ritz method to free vibration and buckling analysis of highly anisotropic plates. *Composite Structures* **2018**, *192*, 460–474.
45. Kuehn, T.; Pasternak, H.; Mittelstedt, C. Local buckling of shear-deformable laminated composite beams with arbitrary cross-sections using discrete plate analysis. *Composite Structures* **2014**, *113*, 236–248.
46. Kharghani, N.; Mittelstedt, C. Global–local–distortional buckling of shear-deformable composite beams with open cross-sections using a novel GBT–Ritz approach. *Journal of Composites Science* **2025**, *9*, 608. https://doi.org/10.3390/jcs9110608
47. Kharghani, N.; Mittelstedt, C. Buckling analysis of composite box girders using generalized beam theory (GBT) and Ritz method. In *Engineering Materials, Structures, Systems and Methods for a More Sustainable Future*; CRC Press: London, UK, **2025**; pp. 224–229. https://doi.org/10.1201/9781003677895-37
48. Kharghani, N.; Mittelstedt, C. Generalized beam theory and Ritz method for buckling analysis of thin-walled composite beams with unsymmetric layups. *ce/papers* **2026**, *9*, 791–796. https://doi.org/10.1002/cepa.70366
49. Kharghani, N.; Mittelstedt, C. Efficient buckling analysis of thin-walled composite beams with symmetric and unsymmetric layups using a GBT–Ritz approach. *Journal of Composites Science* **2026**, *10*, 307. https://doi.org/10.3390/jcs10060307
50. Mindlin, R.D. Influence of rotatory inertia and shear on flexural motions of isotropic, elastic plates. *Journal of Applied Mechanics* **1951**, *18*, 31–38. https://doi.org/10.1115/1.4010217
51. Whitney, J.M.; Pagano, N.J. Shear deformation in heterogeneous anisotropic plates. *Journal of Applied Mechanics* **1970**, *37*, 1031–1036. https://doi.org/10.1115/1.3408654
52. Reddy, J.N. *Mechanics of Laminated Composite Plates and Shells: Theory and Analysis*, 2nd ed.; CRC Press: Boca Raton, FL, USA, **2004**.
53. Vlasov, V.Z. *Thin-Walled Elastic Beams*; Israel Program for Scientific Translations: Jerusalem, Israel, **1961**.
54. Lehoucq, R.B.; Sorensen, D.C.; Yang, C. *ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*; SIAM: Philadelphia, PA, USA, **1998**.